Robotics Foundation Models

Trajectories and open problems

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About Me

01 · ACADEMIC

Korea Univ · NAVER AI Lab

Multimodal learning · visual reasoning · long-tail generalization. PhD track and industry research.

02 · INDUSTRY

maum.ai · Head of Research

Leading the research org for physical-world foundation models. Full stack: data · policy · world models.

03 · WORV

World Robot Verse

maum.ai's physical-AI division. Returns in Ch8 — both the origin and the conclusion of today's story.

Today

8 chapters · 77 slides · ~70 min

01

Robotics Foundation Models?

What does "foundation" actually mean in robotics?

Two robotics summers, two failures

SUMMER #1 · 2015 — 2020

Classical / RL

DeepMind **QT-Opt** · OpenAI **Rubik's Cube** (one-handed). Massive simulation farms + on-policy RL.

Limit: sim-to-real gap, no task transfer.

SUMMER #2 · 2022 — PRESENT

Imitation / Foundation

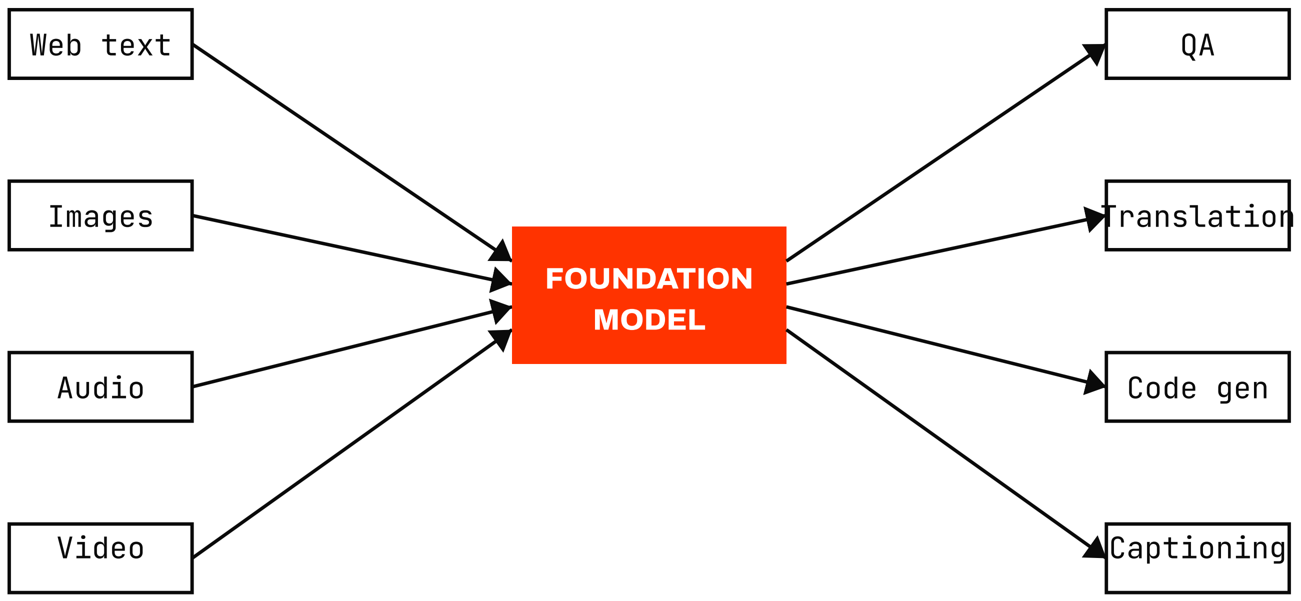
Google **RT-1** · Stanford **Mobile ALOHA**. Human demonstrations + transformer scaling.

Limit: data volume, embodiment transfer, long-horizon — all open.

We're now in the third summer. The difference is the explicit introduction of foundation — one model that generalizes across robots · tasks · environments.

What "foundation" means in NLP / Vision

BOMMASANI ET AL. · STANFORD CRFM · 2021

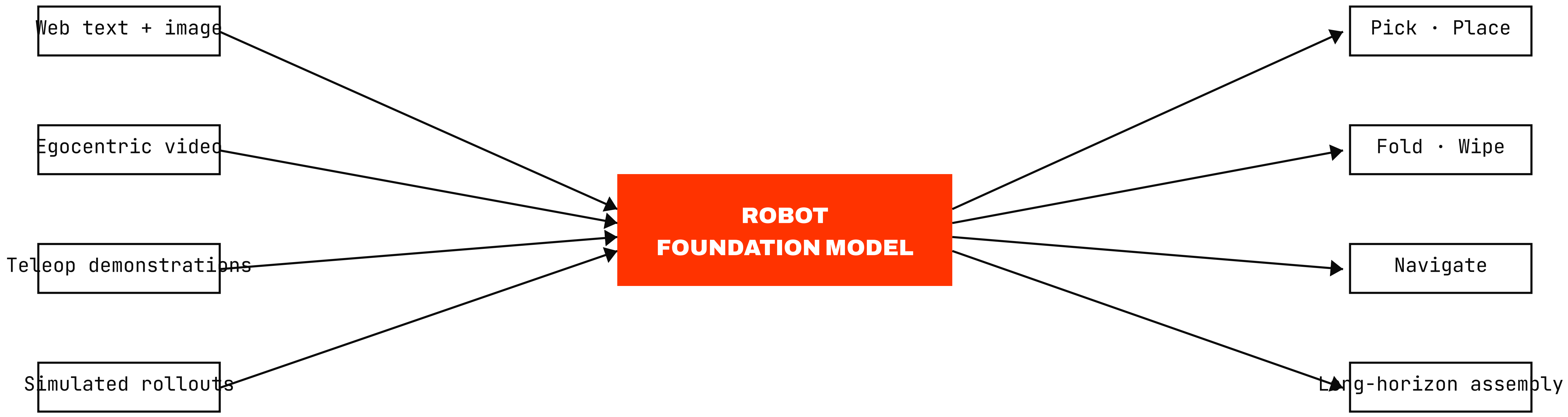


One model → many downstream tasks

- Large-scale self-supervised pretraining
- Downstream tasks need only light fine-tuning / zero-shot
- Capability emergence backed by scaling laws

Already proven in NLP / Vision — next slide: *what breaks when you copy this to robotics?*

What "foundation" means for robots



Input: multimodal observation (vision + proprioception + language) **Output:** physical action — both must generalize

Three axes of generalization

AXIS 1

Task

New verbs in a known environment. *"fold the towel" vs "roll the towel".*

flagship — RT-2 (semantic reasoning)

AXIS 2

Environment

Same task, unseen rooms / lighting / objects. *π0.5 reaches 94% follow-rate in unseen homes.*

flagship — π0.5 (OOD home)

AXIS 3

Embodiment

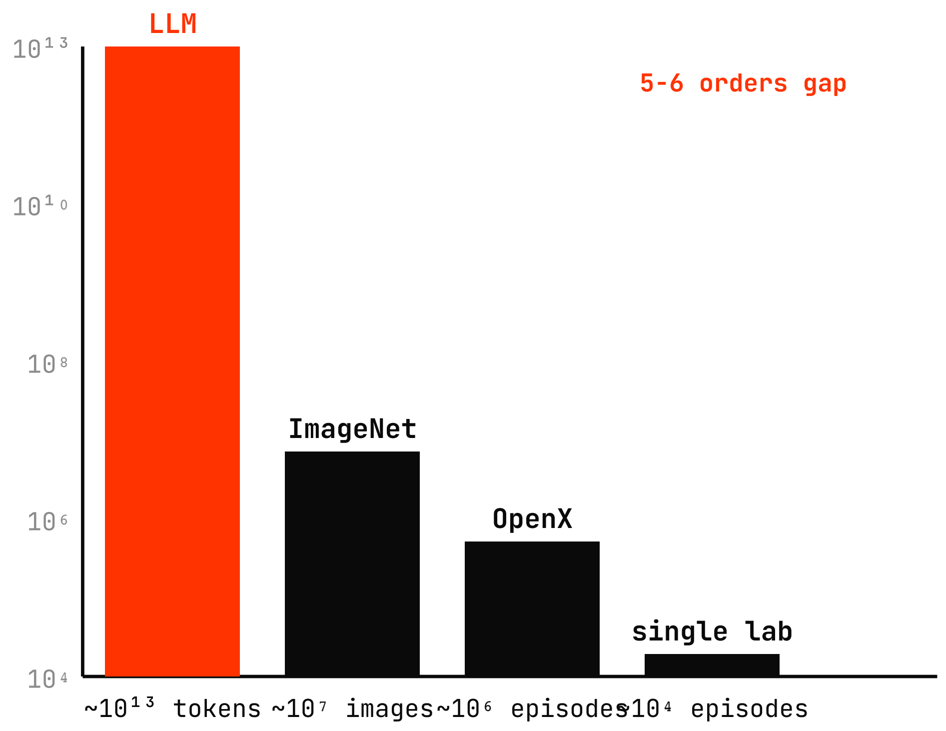
Brand-new hardware, zero-shot. *DreamZero adapts to a YAM arm with 30 minutes of plays.*

flagship — DreamZero · GR00T N1

Most recent papers bet on **one** of these three axes. No model generalizes across all three yet — that's the destination of "robotics foundation."

Why "scaling" alone isn't obvious here

DATA SCALE COMPARISON · LOG AXIS



"We don't have an internet of actions."

Unlike internet text, **action data doesn't grow on its own**. RT-1 ~130k episodes vs LLM ~trillions of tokens — a 5-6 order-of-magnitude gap.

So we need *data tricks* to scale — Ch3 is entirely about this problem.

The 2023 — 2026 unlock

2022	2023	2024	2024-10	2025-02	2025-03	2025-04	2026-02
RT-1	**RT-2**	**OpenVLA**	**π0**	**Helix**	**GR00T N1**	**π0.5**	**DreamZero**
130k episodes, discretized actions	web knowledge → actions	7B open, OpenX 970k	flow-matching action expert	S2 7B + S1 80M @ 200 Hz	open humanoid stack	unseen home 94% follow	joint video + action generation

> Three years on one page — **action tokenization → web knowledge transfer → open stack → dual system → OOD generalization → joint video × action generation**.

arXiv: 2212.06817 · 2307.15818 · 2406.09246 · 2410.24164 · 2503.14734 · 2504.16054 · 2602.15922

Industry stack 2026 H1

🇰🇷 RED BORDER = K-HUMANOID ALLIANCE · WORV @ MAUM.AI

FIGURE
F.03
live stream · humanoid

GENERALIST
GEN-1
multi-task demo

GENESIS
GENE-26.5
manipulation

NVIDIA
Alpamayo
AV foundation

PHYSICAL INTELLIGENCE
π0.5
OOD home · 94%

SKILD AI
Brain
$1.4B unicorn

APPTRONIK × GOOGLE
Apollo
Gemini Robotics

BOSTON DYNAMICS
Atlas
electric · CES'26

1X
NEO
consumer humanoid

TESLA
Optimus
factory deploy

SANCTUARY × MAGNA
Phoenix
dexterous

AGIBOT
G2
first factory line

UNITREE
H1 · G1
Spring 679M views

FOURIER
GR-2
humanoid

GALBOT
G1
wheeled humanoid

GALAXEA
G0 VLA
VLA stack

🇰🇷 NAVER LABS
Ambidex
bimanual research

🇰🇷 RAINBOW
RB-Y1
wheeled humanoid '24

🇰🇷 DOOSAN

🇰🇷 HYUNDAI × BD

🇰🇷 MAUM.AI

🇰🇷 HOLIDAY

🇰🇷 ROBOTIS

🇰🇷 YUJIN

Industry-led — and Korea is part of it · multiple K-Humanoid Alliance members

Roadmap — 7 questions, 7 chapters

CH 2	Action modality	How do we tokenize action?
CH 3	Data scaling	How do we collect and grow that action data?
CH 4	VLAs	How does a VLM become a VLA?
CH 5	Diffusion + Dual	Speed and reasoning, simultaneously
CH 6	World Models	Can a model learn the world's dynamics?
CH 7	World Action Models	Does "knowing" turn into zero-shot policies?
CH 8	WoRV @ maum.ai	How we're working on this from Korea — and who to talk to about joining

02

A New Modality: Action

Text, image, audio — you already know how to tokenize. What about action?

Recap — the modalities you already know

WEEK 10 DECK · MULTIMODAL LLM FUSION PATTERNS

MODALITY	CANONICAL TOKENIZER	EXAMPLE MODEL
Text	BPE / SentencePiece	GPT, Llama
Image	ViT patches / VQ-VAE	CLIP, LLaVA, Chameleon
Audio	Mel-spec / EnCodec	Whisper, AudioLM
Video	Tubelets / latent frames	VideoMAE, Sora
ACTION	**— today's question —**	RT-1, π0, DreamZero

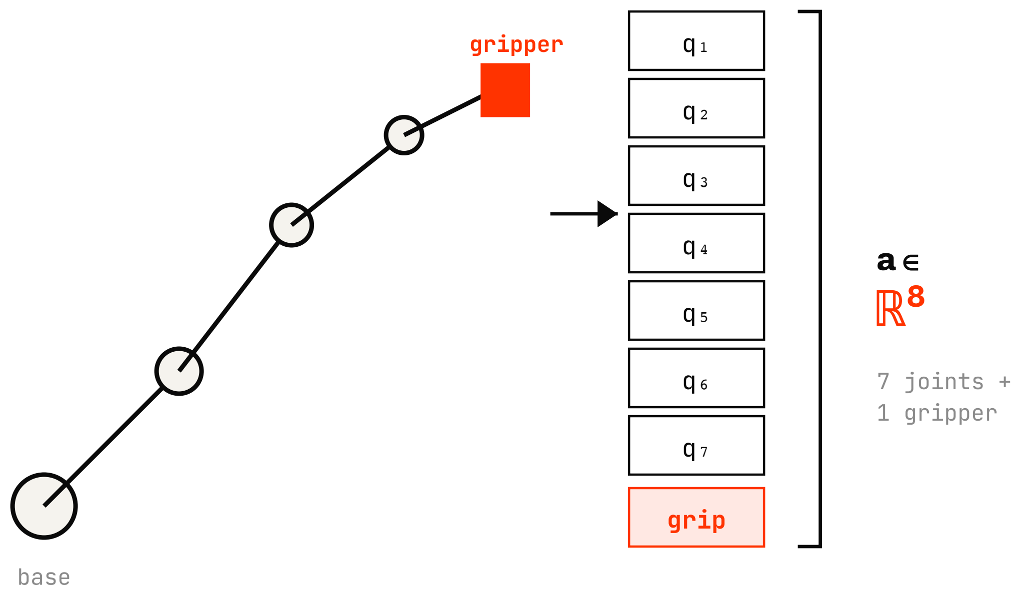
CLIP · BRIAN ICHTER @ STANFORD SEMINAR

"NATURAL OVERLAP" BETWEEN TEXT TOKENS AND ACTION TOKENS · 06:36–08:00

We are *adding one more row* to a table the Week 10 deck already filled in.

What *is* an "action" in a robot?

ONE CONTROL STEP · 7-DOF ARM + 1 GRIPPER



A continuous vector — but embodiment-bound

- **Joint angles** (Franka) *vs* **end-effector pose** (RT-1) *vs* **motor torques** (Atlas)
- **7-DoF** single arm → **14-DoF** bimanual ALOHA → **35-DoF** Helix upper body
- Grippers add a 1-D switch; dexterous hands add 20+ extra DoF
- The "action space" is *not* portable — every robot speaks its own language

Looks like a simple continuous vector. Hides *embodiment dependence* — the bill we pay later.

Why action is harder than text

1 · DISCRETENESS

Text: discrete by birth. Action: continuous.

Cross-entropy "just works" on 50k tokens. Joint angles live in $\mathbb{R}^n$; we must invent the alphabet.

2 · DETERMINISM

Text: next-token softmax. Action: real physics.

The same command on the same scene yields a different outcome — mass, friction, contact, noise.

3 · DATA SOURCE

Text: scrape the internet. Action: teleop, one hour at a time.

10^{13} tokens vs 10^{6} teleop episodes — *no internet for tying shoelaces*.

4 · REVERSIBILITY

Text: regenerate the answer. Action: drop the cup, it breaks.

Mistakes have physical cost — safety, supervision, evaluation all become first-class problems.

CLIP · CHELSEA FINN @ YC AI STARTUP SCHOOL

"NO INTERNET FOR ROBOT ACTIONS" · 02:08–03:32

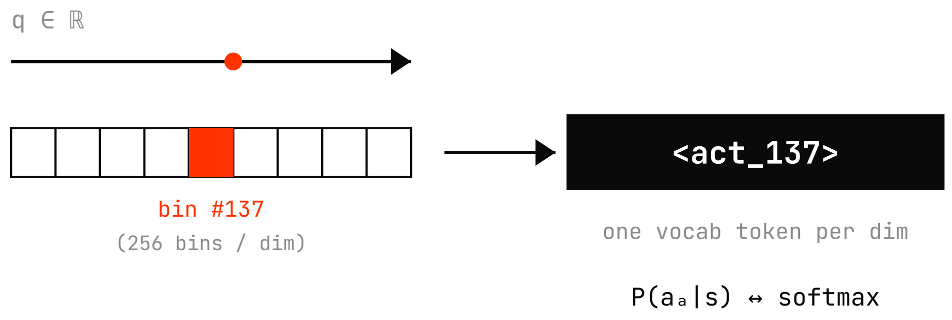
> The whole rest of the talk is a response to **cell #3**. Ch3 fixes data; Ch5 fixes speed; Ch6–7 fix evaluation.

Two tokenization strategies

STRATEGY A

Bin discretization

Slice each continuous dim into **256 bins**, treat each bin as a vocab token, predict with cross-entropy — same loop as an LLM.



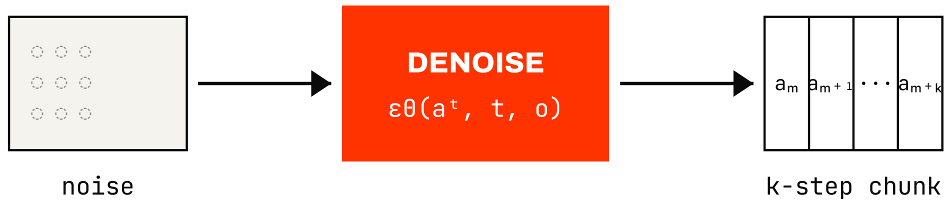
- Reuses LLM loss, optimizer, decoder verbatim
- Quantization error; per-dim factorization

flagship — RT-1, RT-2, OpenVLA

STRATEGY B

Continuous head

Keep the action continuous; bolt a separate **diffusion / flow** head on top of the VLM that denoises a whole action *chunk* at once.



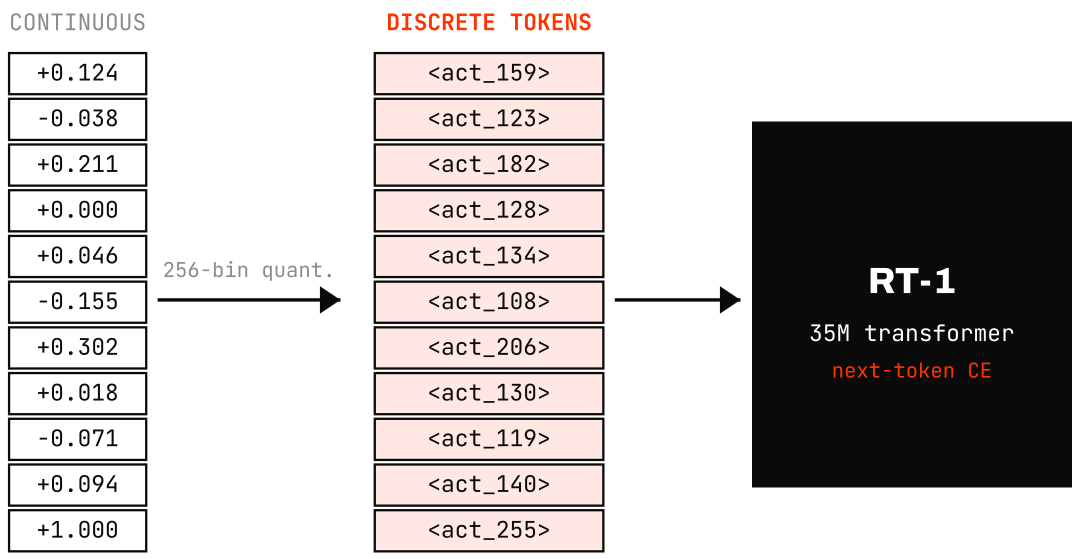
- Captures multimodal action distributions (no mode collapse)
- Slower per call — but predicts a whole chunk at once

flagship — Diffusion Policy, ACT, π0

This is the **#1 design choice** behind every modern VLA. The next three slides take one example of each — and a 2025 hybrid that re-discretizes.

Discrete — RT-1's action vocabulary

11 DIMS × 256 BINS PER DIM



CLIP · RT-1 SUPPLEMENTARY VIDEO

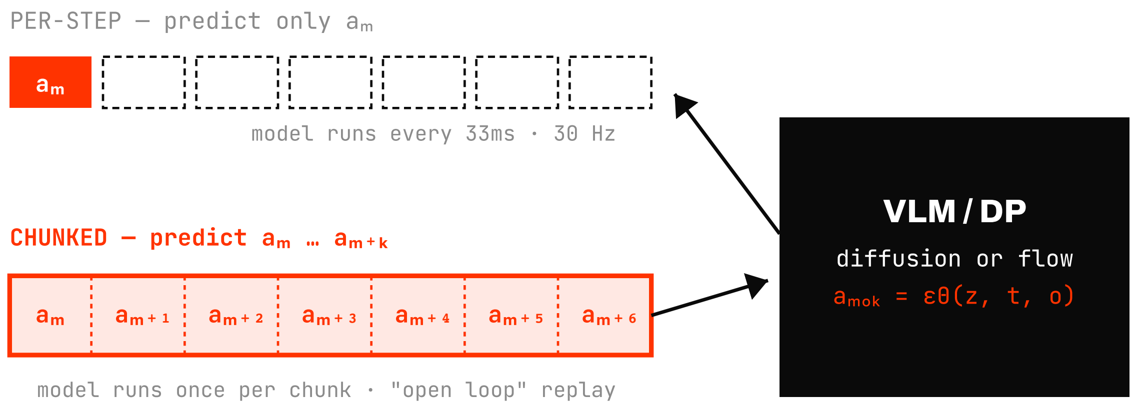
EFFICIENTNET → TOKENS → ACTION TOKENS · 02:07–05:17

- **11 dims** = 7 arm joints + 3 base velocities + 1 gripper
- **256 bins** per dim → 11 tokens emitted per control step
- Loss = next-token cross-entropy — *identical* to an LLM
- Opens the door to VLM → **VLA** (Ch4)

[ARXIV:2212.06817] · Brohan et al. "RT-1: Robotics Transformer for Real-World Control at Scale"

Continuous — action chunks

PREDICT K FUTURE ACTIONS IN ONE SHOT



- **ACT** — bimanual VAE-Transformer trained with action-chunk MSE
- **Diffusion Policy** — same DDPM you saw in Week 12, but the target is an action sequence, not pixels
- Chunking absorbs human teleop noise + handles multimodal action distributions

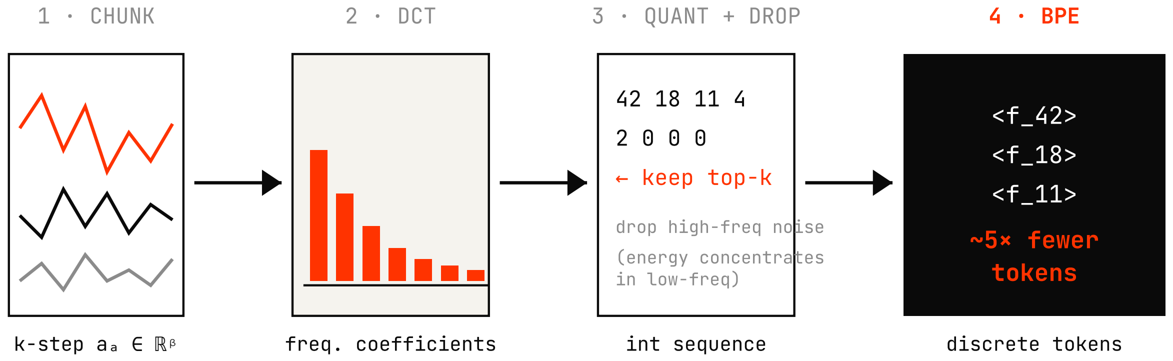
CLIP · RUSS TEDRAKE @ PRINCETON ROBOTICS SEMINAR

"DIFFUSION POLICY = SEQUENCE OF FUTURE ACTIONS" · 20:02–21:05

DDPM → policy. Same noise schedule, same εΘ objective. Only the target tensor changes — from pixels to an *action chunk*.

FAST — a compression-based tokenizer

Π0-FAST · PERTSCH · JAN 2025



- **Continuous → back to discrete**, but each token carries more info
- Energy concentrates in low-frequency — *drop the noise tail for free*
- **5× shorter** sequences than RT-1-style binning → faster train, faster inference

CLIP · SERGEY LEVINE @ TWIML #719

"A BETTER WAY TO COMPRESS ACTIONS" — FAST MOTIVATION · 39:17–40:25

> The third option: **continuous → back to discrete**, but in a basis where each token carries real information.

Takeaway — action is the new token

SAME TABLE AS SLIDE 14 — WITH ACTION FILLED IN

MODALITY	CANONICAL TOKENIZER	EXAMPLE MODEL
Text	BPE / SentencePiece	GPT, Llama
Image	ViT patches / VQ-VAE	CLIP, LLaVA
Audio	Mel-spec / EnCodec	Whisper
Video	Tubelets / latent frames	Sora
ACTION	**(A)** Bin discretization · 256/dim	RT-1, RT-2, OpenVLA
	(B) Continuous chunk + diffusion / flow	ACT, Diffusion Policy, π0
	(C) FAST · DCT → BPE re-discretize	π0-FAST

> Action is now a modality that **LM**, **diffusion**, and **flow** can all consume. Next chapter — where do we get the *data*?

CLOSER CLIP · FEI XIA @ STANFORD CS25 V3

"ACTION AS YET ANOTHER LANGUAGE" · 58:12–59:25

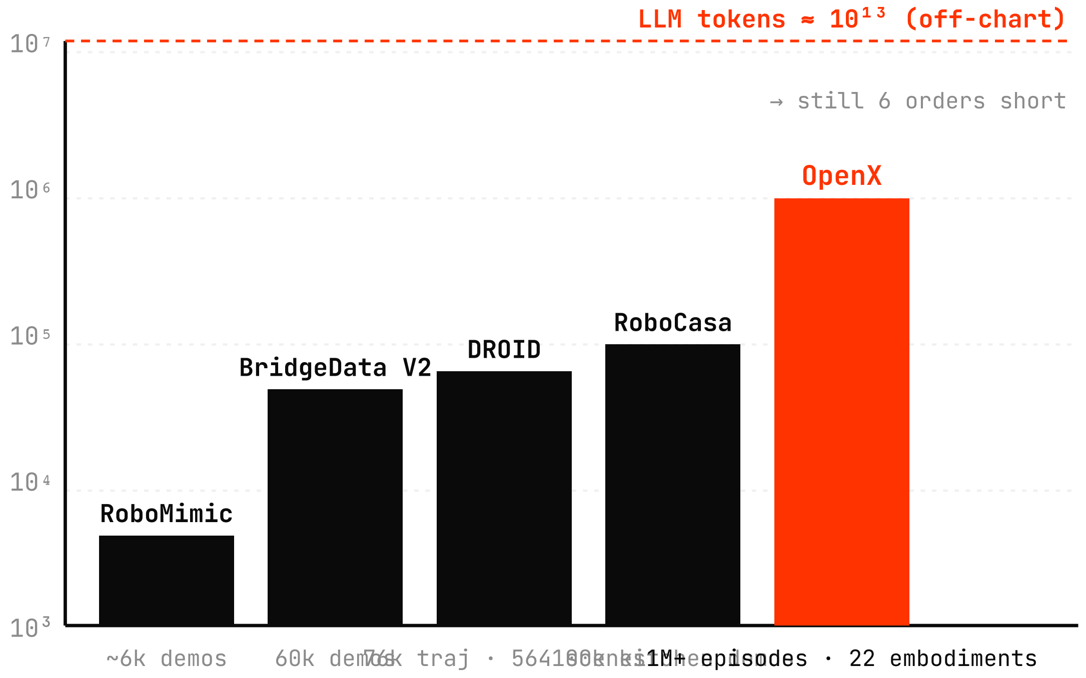
03

Data Scaling.

If actions don't grow on the internet, how do we scale them?

The action data wall

ROBOT-LEARNING DATASETS · LOG SCALE



5-6 orders of magnitude short.

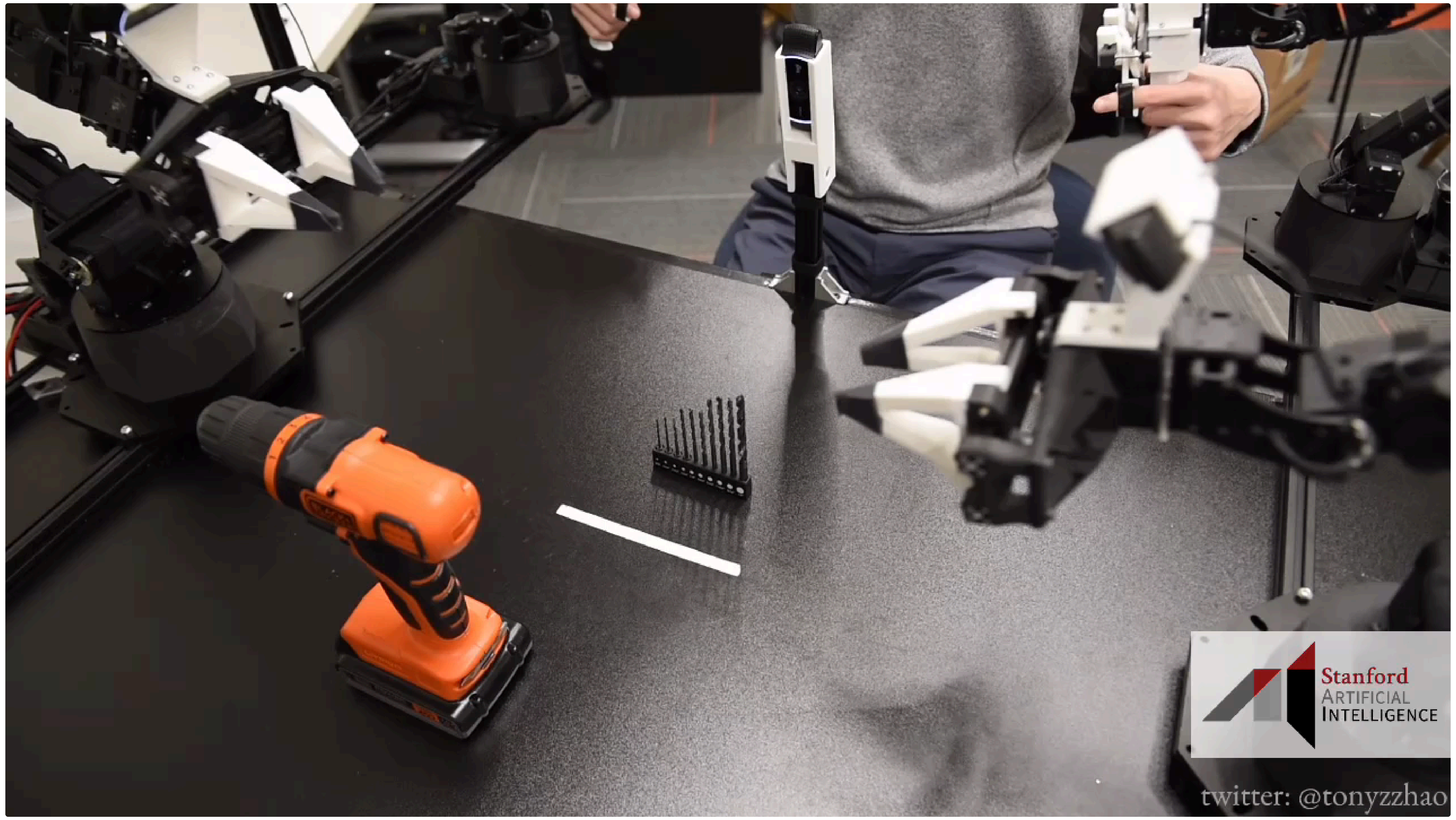
- Every **jump on this chart** = a new collection method
- Lab teleop < cross-lab union < in-the-wild < human video < sim
- The rest of the chapter = four strategies to close the gap

Plus OpenX-Embodiment as the *union* of nearly everyone's lab data — see slide 29.

arXiv: 2108.03298 · 2308.12952 · 2403.12945 · 2406.02523 · 2310.08864

Strategy 1 — Better hardware (ALOHA family)

CHEAPER TELEOP → MORE DATA



ALOHA BIMANUAL TELEOP HERO — 6 DEXTEROUS TASKS BACK-TO-BACK · ZHAO ET AL. 2023

ALOHA
2023 · Stanford

Open-source bimanual leader-follower rig, **under $20k**. Paired with ACT (Action Chunking Transformer).

Mobile ALOHA
2024 · Stanford

Wheeled base + ALOHA arms → **whole-home tasks** (cooking, laundry, elevator).

ALOHA 2 · ALOHA Unleashed
2024 · DeepMind

v2 hardware + sim · **Unleashed = diffusion + lots of teleop** → shoelace tying, gear insertion.

price ↓ = data ↑ · cheap rigs let *any* lab contribute.

arXiv: 2304.13705 · 2401.02117 · 2405.02292 · 2410.13126

Strategy 2 — Handheld in-the-wild (UMI → Sunday)

NO ROBOT NEEDED AT COLLECTION TIME

UMI

[ARXIV:2402.10329]

Chi et al. · RSS 2024

GoPro + handheld gripper → **robot-compatible data collected anywhere** (home, restaurant, outdoors). Cuts the per-episode cost by an order of magnitude.

Sunday Glove · Memo

[LAUNCH]

Sunday Robotics · Nov 2025

Same UMI team productized it: a **$200 wearable glove** + Memo humanoid. 2,000+ gloves shipped, 10M household-chore episodes from 500 homes.

Narrative bridge — **Tony Zhao + Cheng Chi** (ALOHA · Mobile ALOHA) → spun out as Sunday. Same people, lower friction at every step: $20k rig → $400 GoPro rig → $200 glove.

Strategy 3 — Egocentric video (Ego4D · Ego-Exo4D)

MASSIVE · UNLABELED · FIRST-PERSON

Ego4D

[ARXIV:2110.07058]

Grauman et al. · CVPR 2022

3,670 hours of first-person video from 923 participants across 74 cities, 9 countries. The biggest single ego-video pool.

Ego-Exo4D

[ARXIV:2311.18259]

Grauman et al. · CVPR 2024

Paired ego + exo third-person views of the same activity — supplies the alignment needed for body / hand transfer.

> Human video has **no action labels** — but it is the only pool that already exists at scale. The next slide (EgoMimic / EgoVLA / HumanPlus) is about *how* we turn pixels into policy. Forward-ref → Ch7 LAPA recovers latent actions directly from these frames.

Strategy 3+ — Video → robot

EMBODIMENT BRIDGE, NOT JUST SUPERVISION

EgoMimic

Georgia Tech · ICRA 2025

Aria glasses + bimanual robot, **co-trained** on paired human + robot data.

[ARXIV:2410.24221]

EgoVLA

HUMAN-VIDEO → VLA

short + long horizon
sort-cans · stack-in-drawer

EgoVLA

NVIDIA + UCSD · 2025

Pretrain a VLA on **human video**, fine-tune on a tiny robot set.

[ARXIV:2507.12440]

HumanPlus

Stanford · CoRL 2024

Unitree H1 **shadows** humans from 3rd-person video. Boxing, piano, table tennis.

[ARXIV:2406.10454]

Not just *more* supervision — an **embodiment bridge**. Human pixels transferred to robot policy, then to humanoid bodies.

Strategy 4 — Synthetic (sim + world-model dreams)

NO HUMANS IN THE LOOP

NVIDIA "FROM DREAMS TO REALITY" — DREAMGEN / GR00T-DREAMS SYNTHETIC TRAJECTORIES

Isaac Lab

GPU-accelerated physics sim. Thousands of parallel envs, randomized scenes / lighting / textures.

RoboCasa

100k procedurally-generated **kitchen tasks** — appliances, layouts, AI textures.

DreamGen · GR00T-Dreams

[ARXIV:2505.12705]

World model generates trajectories — **3 months → 36 hours** of human data for GR00T N1.5.

Forward-ref → **Ch6 World Models · Ch7 WAMs** reinterpret *generation* itself as the data factory.

OpenX-Embodiment — the union

22 EMBODIMENTS · 60+ LABS · 1M+ EPISODES

OPENX 22-EMBODIMENT TASK MONTAGE — SAME SKILL, DIFFERENT BODIES · COLLABORATION ET AL. 2023

"Let's pool what we have."

Not a new collection method — a **convention** that aligns 60+ labs' formats so they're trainable together.

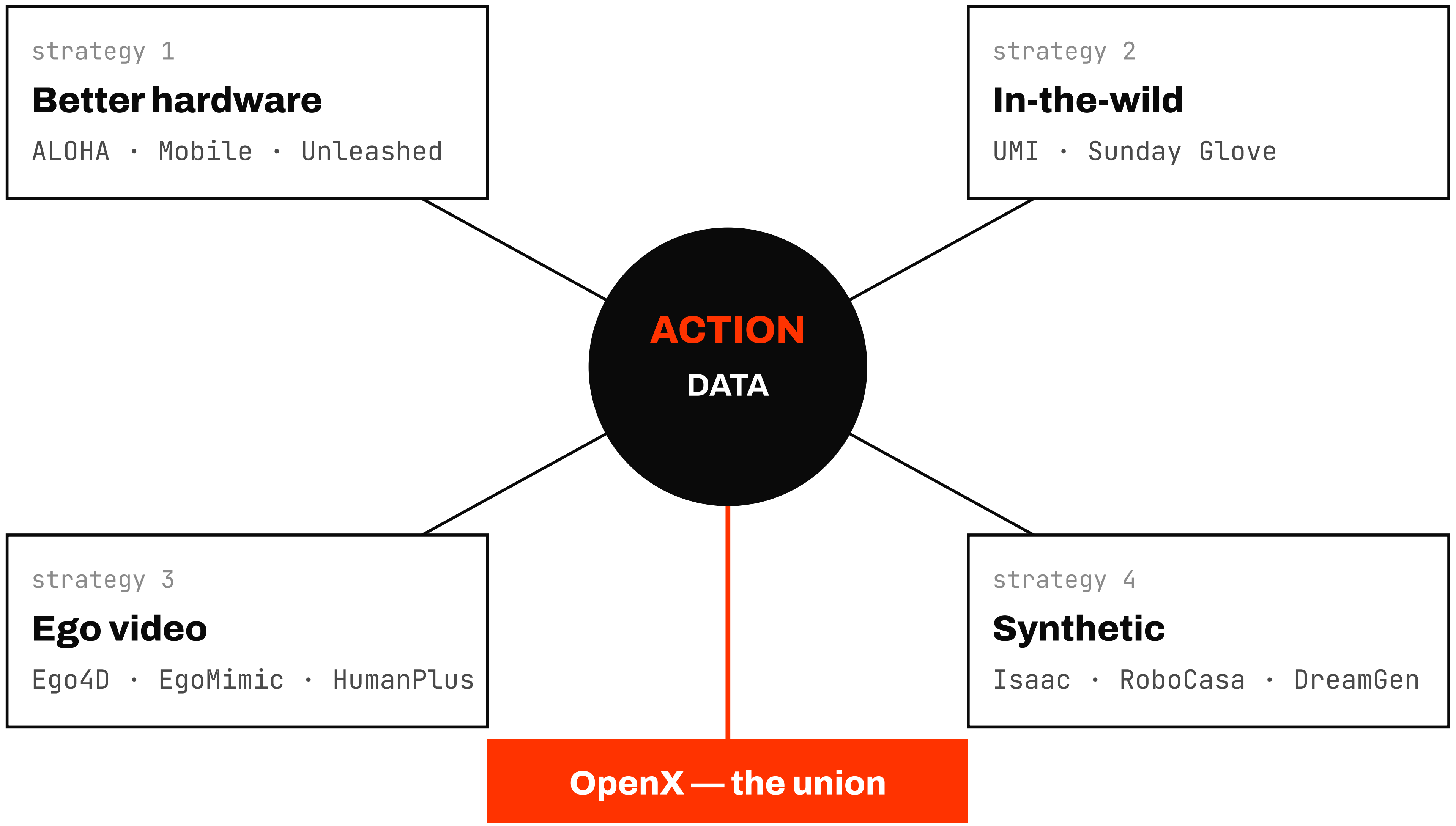
EMBODIMENTS	EPISODES
22	1M+
INSTITUTIONS	**SKILLS**
60+	527

Spawned the dataset wave around it: **DROID** (76k traj, OpenVLA's training pool), **BridgeData V2**, **ALOHA Unleashed**, **RoboCasa**. Same field-wide pressure to grow the pool.

arXiv: 2310.08864 · 2403.12945 · 2308.12952 · 2410.13126 · 2406.02523

Recap — data > model

ALMOST EVERY SOTA JUMP CAME FROM A NEW DATA SOURCE



Almost every SOTA jump came from new data, not a bigger model.

- RT-1 → RT-2 = web data (not bigger ViT)
- OpenVLA = OpenX 970k (not new architecture)
- π0.5 OOD homes = co-training with 22k web episodes
- GR00T N1.5 = DreamGen synthetic (3 mo → 36 hr)

Next chapter — **how do we actually pour this data into a model?**

04

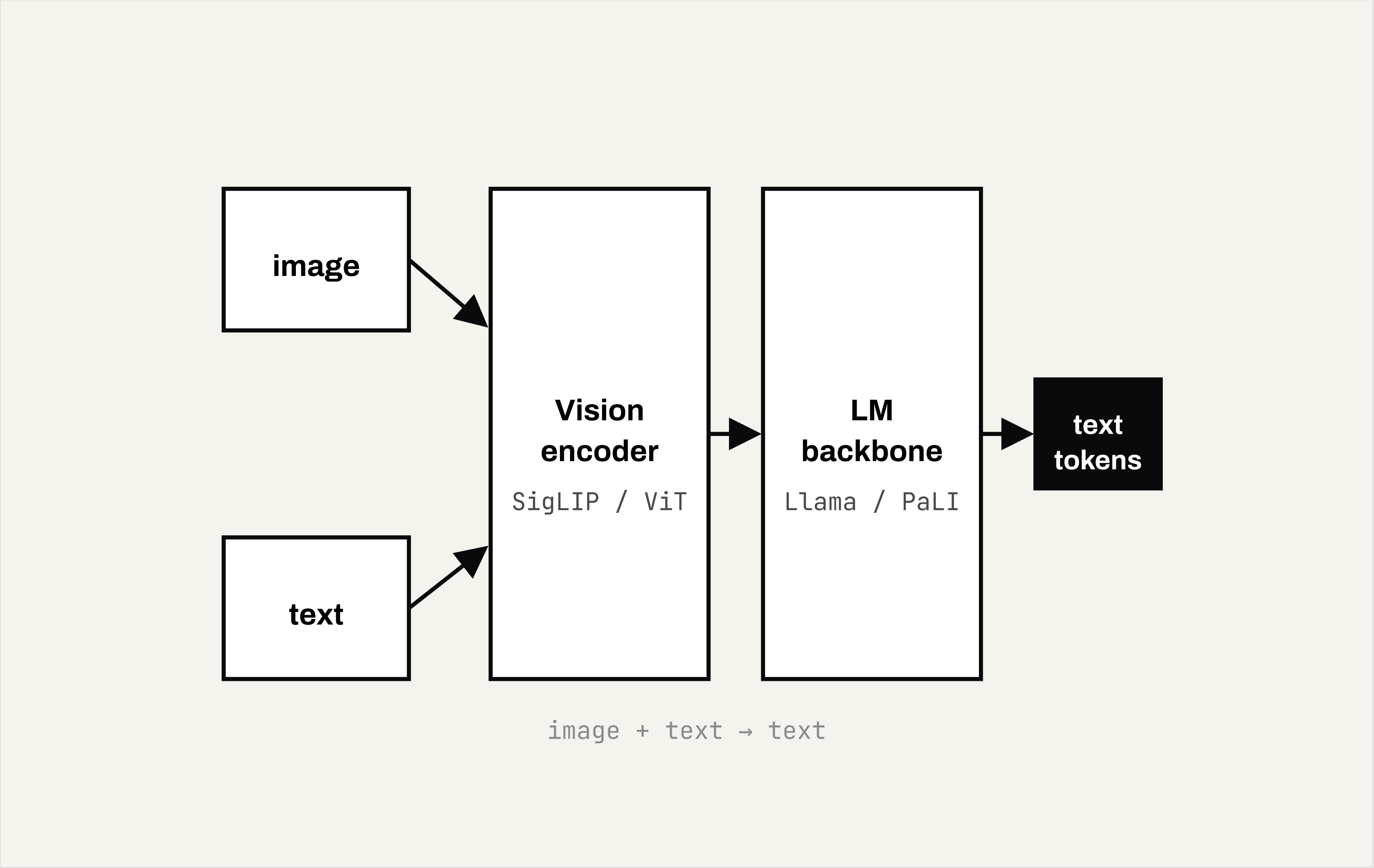
VLAs.

If a VLM outputs text, what does it take to make it output an action?

Definition — swap the head, get a VLA

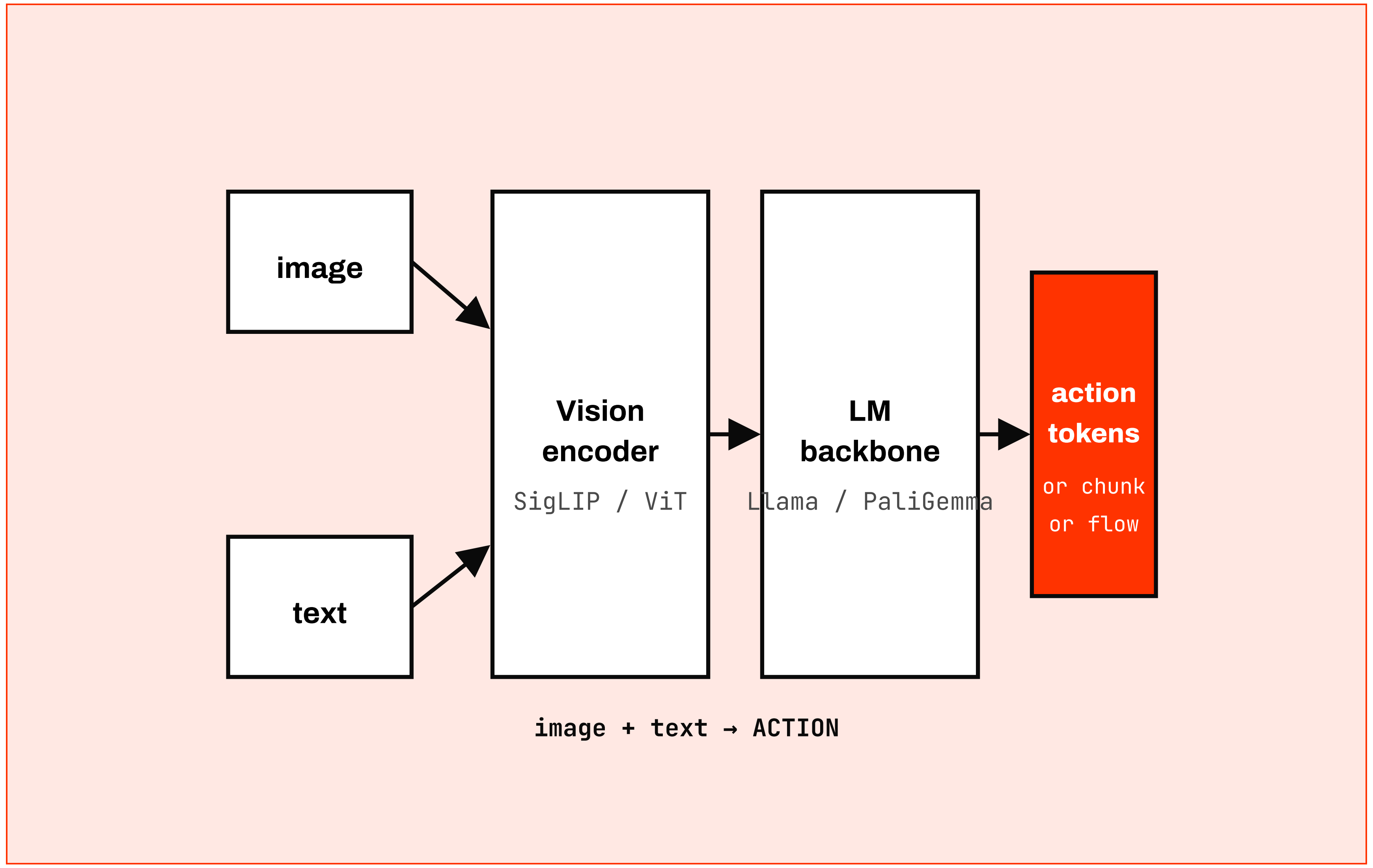
Same vision encoder. Same LM backbone. Different decoder.

A · VISION-LANGUAGE MODEL (VLM)



LLaVA, PaLI, PaliGemma, Chameleon — the row in *last week's* table.

B · VISION-LANGUAGE-ACTION MODEL (VLA)



RT-2, OpenVLA, π0, GR00T — the only structural change is the right-hand block.

VLA ≡ VLM with an action decoder · rest of Ch4 = backbone × head × data choices

RT-2 — web knowledge → robot actions

The paper that made co-fine-tuning on web data + robot data the default.

OFFICIAL · GOOGLE DEEPMIND · "REVOLUTIONIZING ROBOTICS: INTRODUCING RT-2"

ROBOT PICKS THE "EXTINCT ANIMAL" TOY · NEVER TRAINED ON THE WORD "DINOSAUR"

BACKBONE

PaLI-X / PaLM-E

5B · 55B variants, frozen web pre-training kept.

ACTION HEAD

Actions as text tokens

256-bin discretization, integers in the LM's own vocab.

TRICK

Co-fine-tune

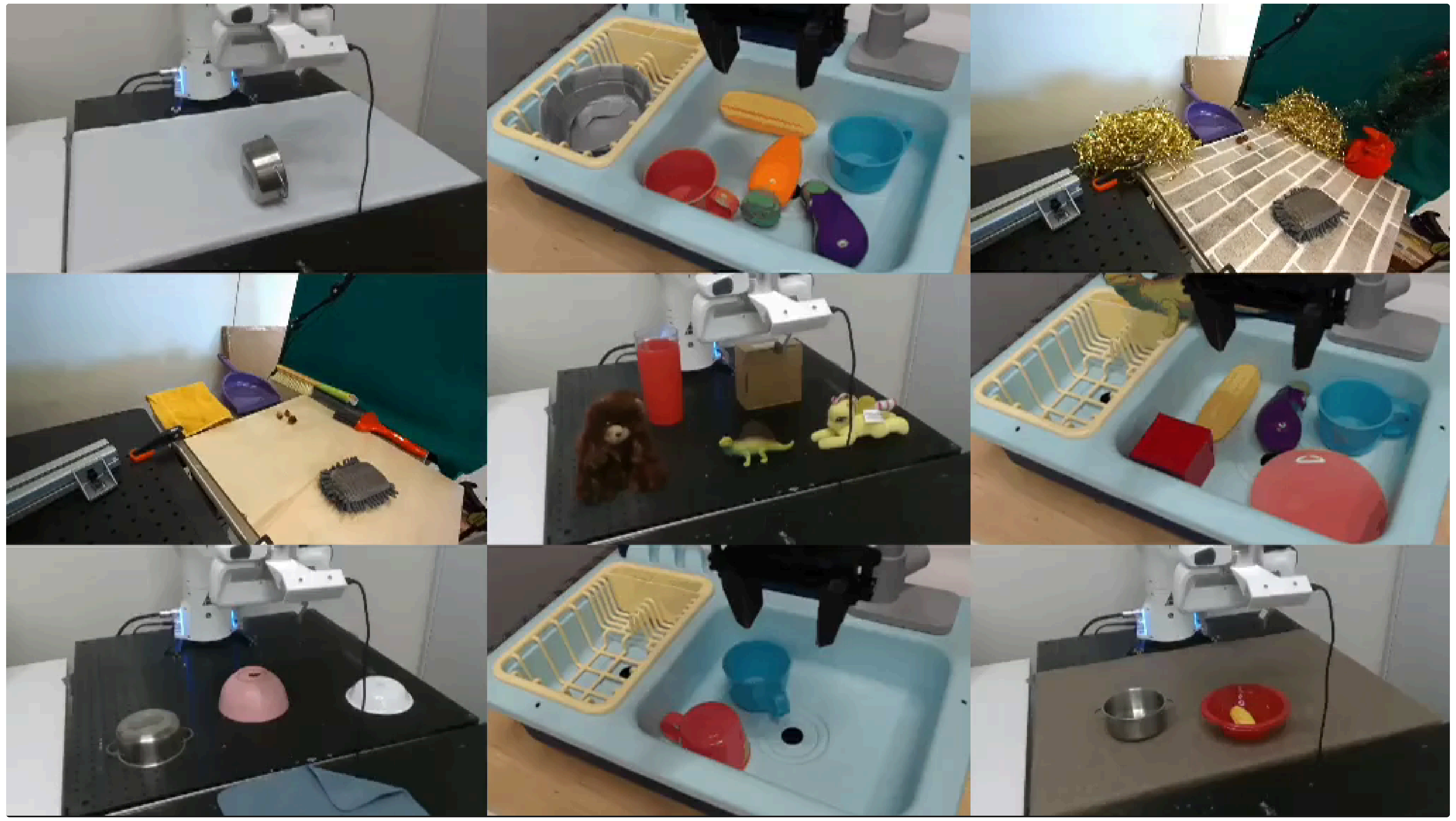
VQA + caption + robot trajectories in one batch → **emergent semantic generalization**.

Result: the robot can act on concepts it never saw in robot data — "*pick the extinct animal*" works because the LM knew the word and the action head spoke the same token language.

OpenVLA — opening the door

The moment academia could reproduce a SOTA VLA on its own GPUs.

PROJECT SITE · OPENVLA.GITHUB.IO



7B PARAMS · LORA FINE-TUNES ON A SINGLE 24 GB GPU · MULTI-EMBODIMENT GENERALIST

7B
PARAMETERS · OPEN WEIGHTS

970k
OPENX EPISODES

RECIPE

SigLIP + DINOv2 → Llama-2 7B → bin tokens

Same head style as RT-2 (256-bin discretization), but every part is open and swap-in/swap-out.

← BRIDGE BACK TO CH3 SLIDE 29

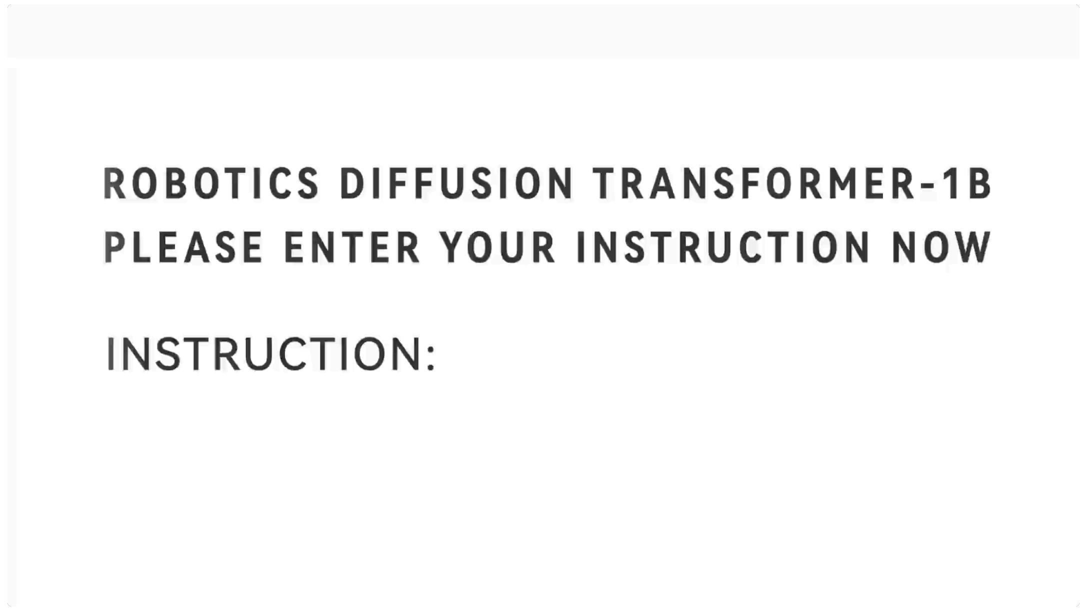
OpenVLA is the **first direct payoff of the OpenX union**. 970k episodes from 22 embodiments — the dataset Ch3 ended on — trained in one pass on one open model.

Generalist policies — three takes on the same year

Same problem, different action head: diffusion vs. block-wise transformer vs. dual-system.

RDT-1B

Oct 2024



Diffusion transformer over bimanual ALOHA. 1B params. Multi-modal action distribution → cleaner mode separation than bin tokens.

Tsinghua · [ARXIV:2410.07864]

Octo

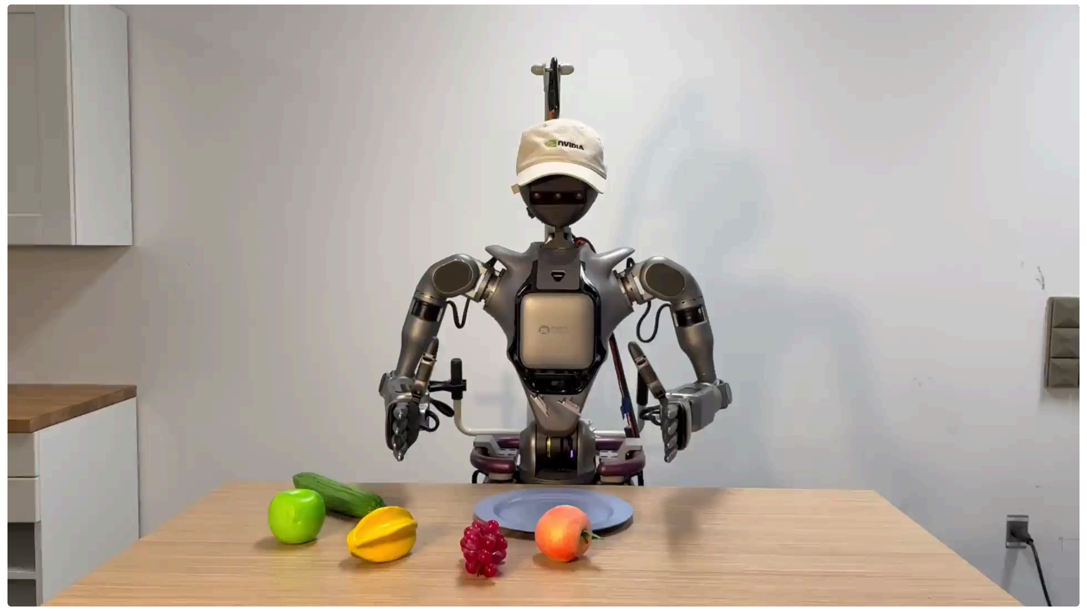
May 2024

Block-wise transformer + diffusion action head. Trained on 800k OpenX trajectories — runs on 9 embodiments out of the box.

Berkeley · [ARXIV:2405.12213]

GR00T N1

Mar 2025



Dual system: Eagle VLM (S2) + DiT action head (S1). First open humanoid foundation model — data pyramid web → human ego → teleop.

NVIDIA GEAR · [ARXIV:2503.14734]

Three labs, twelve months apart, each picking a different action head — **VLA is not a single recipe**. Each design choice (diffusion / block transformer / dual-system) becomes a whole chapter of follow-up work (Ch5).

Open vs closed VLA landscape

2026 H1 SNAPSHOT

	OPEN WEIGHTS	CLOSED
ACADEMIC / OPEN-RESEARCH	★ OPEN + ACADEMIC — "YOUR NEXT SEMESTER" **OpenVLA** '24 · **RDT-1B** '24 · **Octo** '24 · **LAPA** '24 KAIST Weights, code, dataset recipes — LoRA-finetune on one GPU.	CLOSED + ACADEMIC — RARE — *Almost empty cell — if it's academic, the incentive is to release.*
INDUSTRY / PRODUCT	OPEN + INDUSTRY — "OPEN CORE" **GR00T N1** '25 NVIDIA · **GR00T N1.5** '25 NVIDIA · **π0-FAST** '25 PI (partial) Industry labs that release weights for ecosystem leverage.	CLOSED + INDUSTRY — PRODUCT MOAT **RT-1** '22 Google · **RT-2** '23 Google · **π0** '24 PI · **π0.5** '25 PI · **Helix** '25 Figure · **Gemini Rob 1.5** '25 DeepMind · **DreamZero** '26 NVIDIA Paper or blog post + demo videos; weights stay in-house.

12 models · who-publishes-what shapes who-can-research-what

What VLAs are still bad at

VLM limits + robotics-native limits = the four open problems of 2026.

FAIL 01 Precise contact

Threading a needle, plugging a USB, inserting a key — sub-millimeter force-modulated contact is where smooth VLA rollouts fall off a cliff.

`VLM tokens have no haptic channel · tactile sensing not yet in the input pipe.`

FAIL 02 Long horizon

"Make breakfast" — 30 sub-tasks, recovery from a dropped egg, no global plan to fall back on. VLAs drift after ~30s of autonomous rollout.

`No explicit planner · error compounds across token-by-token rollouts.`

FAIL 03 Novel embodiment

Trained on Franka + ALOHA → deployed on a new arm with different DoF, gripper, joint limits. Zero-shot collapse is near-universal.

`Action vocabulary is embodiment-specific · DreamZero needs 30 min of YAM plays to adapt.`

FAIL 04 Speed

RT-2 runs at ~1 Hz; OpenVLA ~5 Hz. Reactive contact and dynamic motion need 30–200 Hz. Big VLM = smart but slow.

`1 Hz vs 200 Hz · the smarter the model, the slower it serves — Ch5's whole motivation.`

Four open problems → Ch5 (faster S1) + Ch6·7 (smarter S2) are direct responses

Two responses — faster S1, smarter S2

The field splits along the Kahneman line. Both halves get their own chapter.

VLA today

smart but slow · contact-fragile · embodiment-locked

↙ SPLIT ALONG KAHNEMAN LINE ↘

S1 Faster motor expert

Take the action head off the LM's critical path. A small, fast expert runs at 30–200 Hz; the VLM only steers it.

- **Diffusion Policy** · denoise an action chunk
- **π0 flow-matching expert** · 50 Hz
- **Helix S1** · 80M params @ 200 Hz
- **FAST tokens** · shorter sequences, same head

→ CHAPTER 5

S2 Smarter high-level brain

Externalize reasoning. Give the VLM a world to roll forward, a plan to follow, a video to imagine.

- **Embodied chain-of-thought** · Gemini Robotics ER 1.5
- **World models as simulators** · rollout-as-reasoning
- **World Action Models** · joint video + action generation
- **DreamZero** · the canonical WAM

→ CHAPTERS 6 & 7

Two halves of the same problem · the rest of today follows one arrow then the other

VLA cheat-sheet — 12 models, one table

BOOKMARK FOR THE REST OF THE TALK

MODEL	YEAR	ORG	BACKBONE	ACTION HEAD	OPEN?
RT-1	2022	Google	FiLM-EfficientNet + Transformer	256-bin discrete tokens (11-dim)	closed
RT-2	2023	Google DeepMind	PaLI-X / PaLM-E (5B·55B)	Actions as text tokens (co-FT)	closed
OpenVLA	2024	Stanford / UW / Toyota	SigLIP + DINOv2 → Llama-2 7B	256-bin discrete tokens	OPEN 7B
Octo	2024	Berkeley / Stanford	Block-wise transformer (93M·27M)	Diffusion action head	OPEN
RDT-1B	2024	Tsinghua TSAIL	DiT-style transformer (1B)	Diffusion over action chunk	OPEN 1B
π0	2024	Physical Intelligence	PaliGemma 3B + 300M expert	Flow-matching action expert	closed
π0.5	2025	Physical Intelligence	Hierarchical π0 (HL + LL VLA)	Flow-matching + HL semantic plan	closed
Helix	2025	Figure AI	S2 VLM 7B (7-9 Hz) + S1 80M (200 Hz)	Latent vector → small motor expert	closed
GR00T N1	2025	NVIDIA GEAR	Eagle VLM (~2B) + DiT	DiT action head (System-1)	OPEN
GR00T N1.5	2025	NVIDIA GEAR	Eagle VLM + DiT + FLARE	DiT + ego-video latent alignment	OPEN
Gemini Robotics 1.5	2025	Google DeepMind	Gemini + ER 1.5 reasoner	Embodied CoT → action	closed
DreamZero (GR00T N2)	2026	NVIDIA GEAR	14B AR video diffusion	Joint video + action generation (WAM)	paper-only

open-weights rows = your candidate list for hands-on work.

Section recap · bookmark for Ch5-7

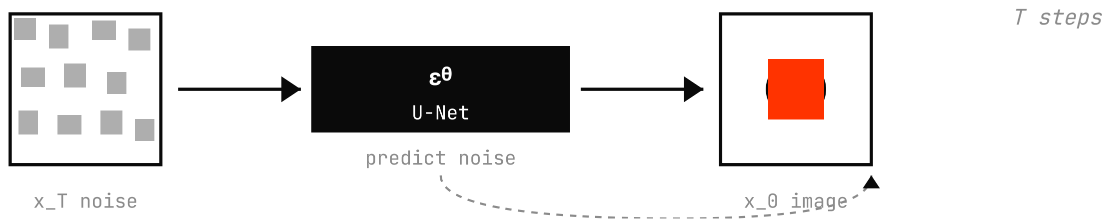
05

Smart & Fast

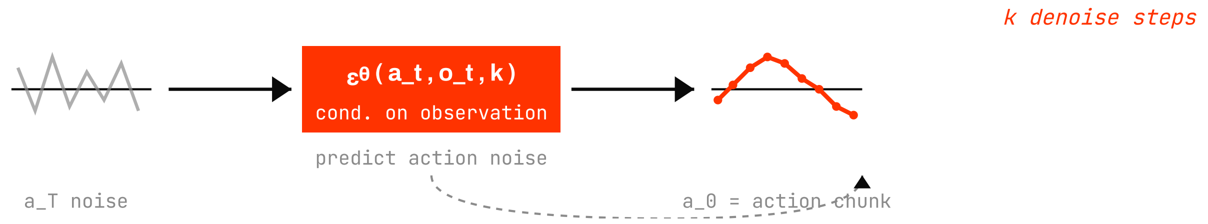
DDPM you just learned, re-cast as a policy — and the dual-system pattern that became consensus in 2026.

Diffusion as policy — from image denoise to action denoise

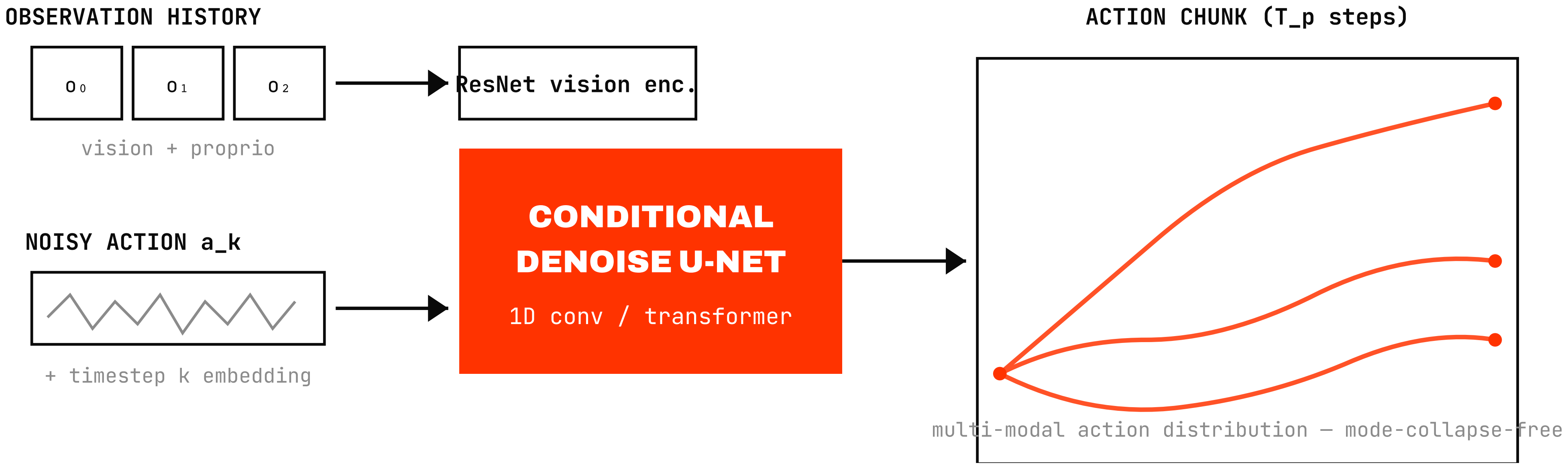
WEEK 12 RECAP · DDPM IMAGE-DENOISE



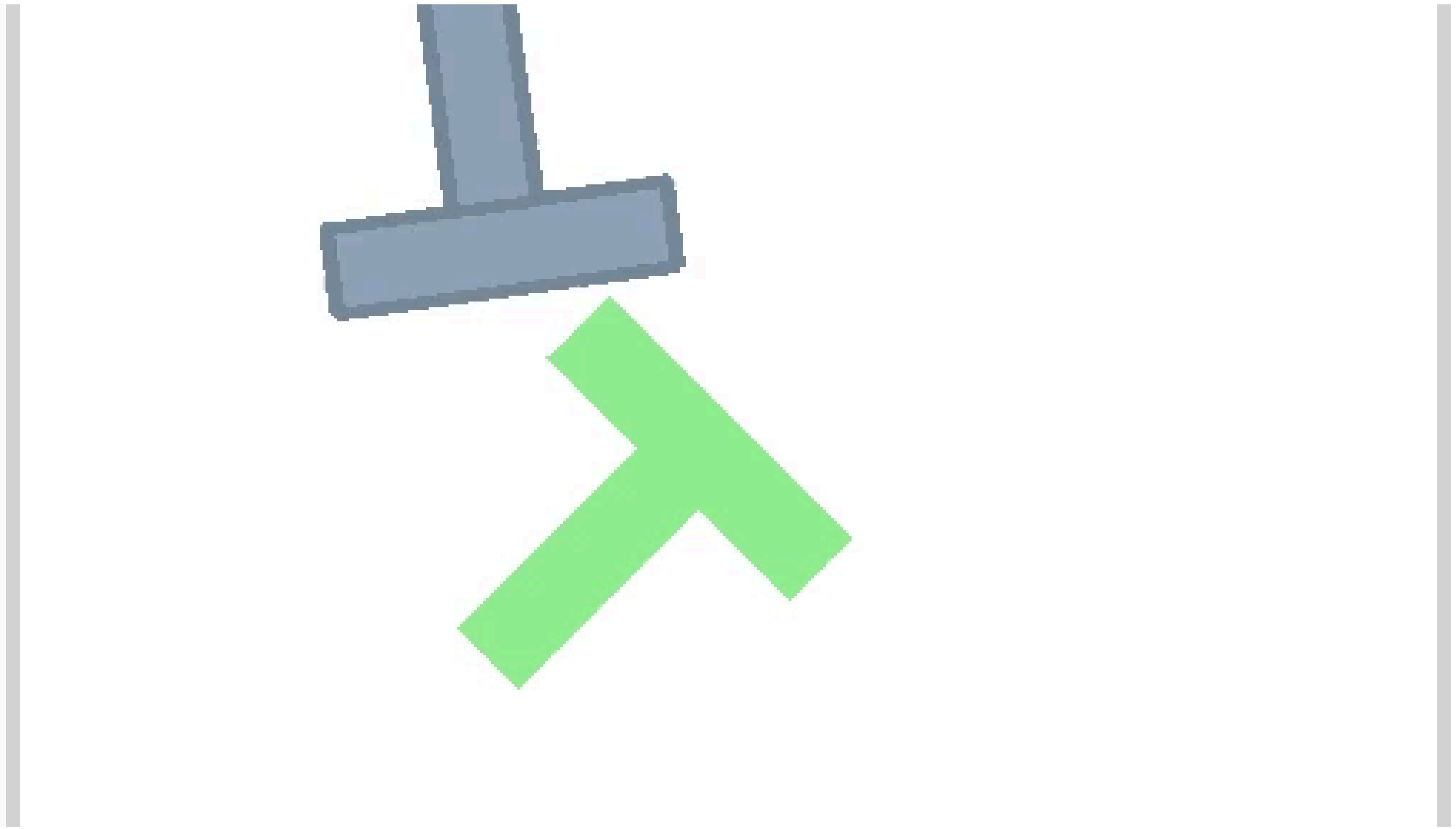
DIFFUSION POLICY · ACTION-DENOISE



CHI ET AL. 2023 · FIG 2 REDRAWN



DEMO · PUSH-T (CHI 2023)

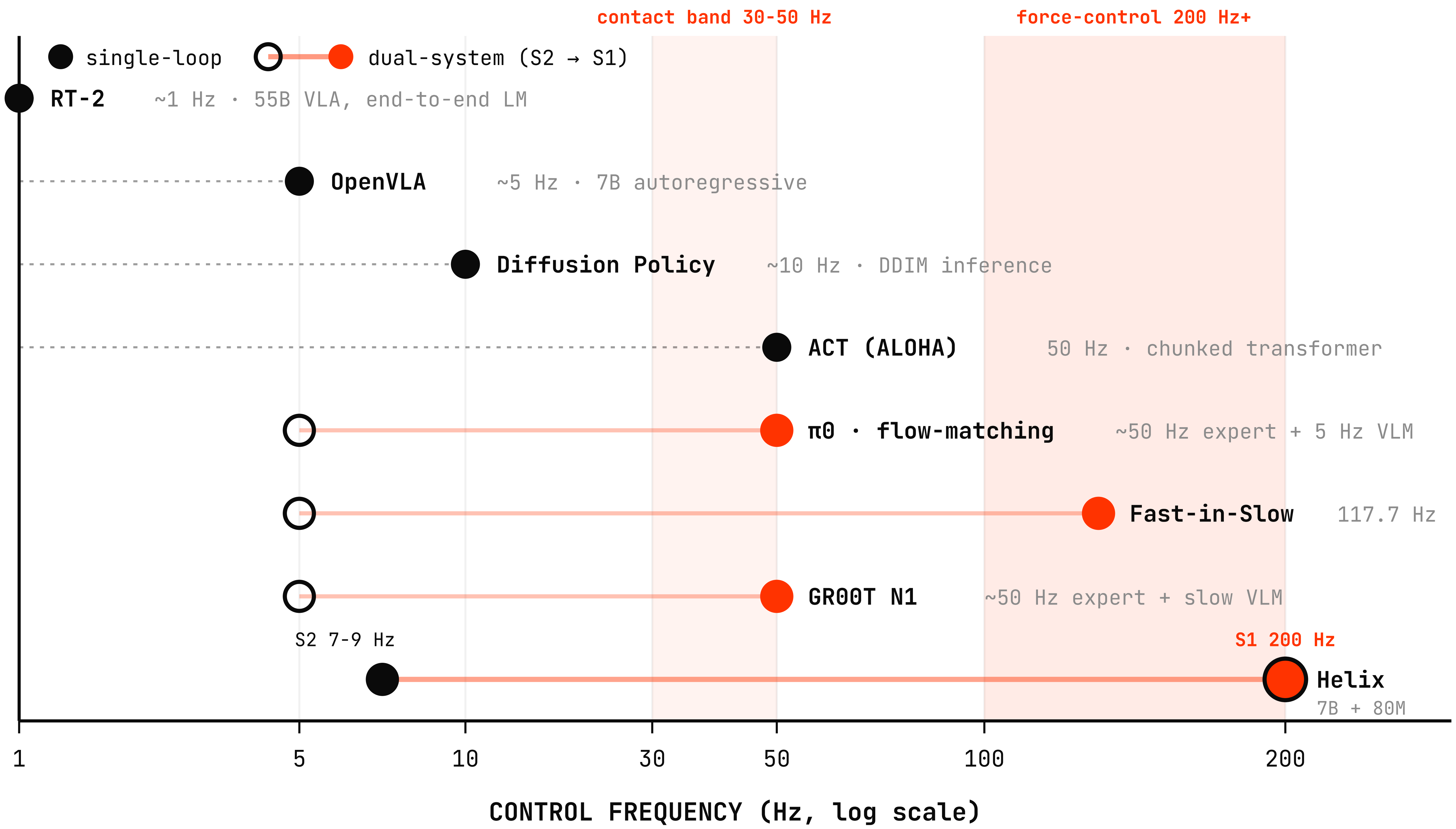


Same ε_θ that denoised pixels now denoises an **action chunk** — conditioned on observations, multi-modal by construction.

Chi, Feng, Du, Xu, Burchfiel, Tedrake, Song · RSS 2023 · [ARXIV:2303.04137] · diffusion-policy.cs.columbia.edu

Why "fast" matters — control frequency

REPORTED INFERENCE / ACTION-LOOP RATE · LOG HZ AXIS



SMARTER = SLOWER

RT-2 has the strongest semantic reasoning — and runs at **1 Hz**. Contact-rich tasks need **30-50 Hz**. Force control needs **200 Hz+**.

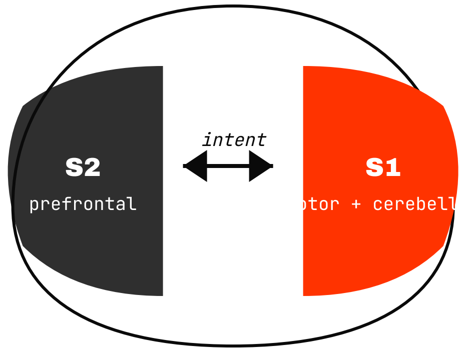
One model can't be both. **Solution:** split into a fast S1 + slow S2 — the chapter's central pattern.

Numbers from each paper's reported inference rate; dashed line = bridge from VLM trigger

RT-2 2307.15818 · OpenVLA 2406.09246 · DP 2303.04137 · ACT 2304.13705 · π0 2410.24164 · Helix figure.ai/news/helix · FiS-VLA 2506.01953 · GR00T N1 2503.14734

Dual system — the Kahneman analogy

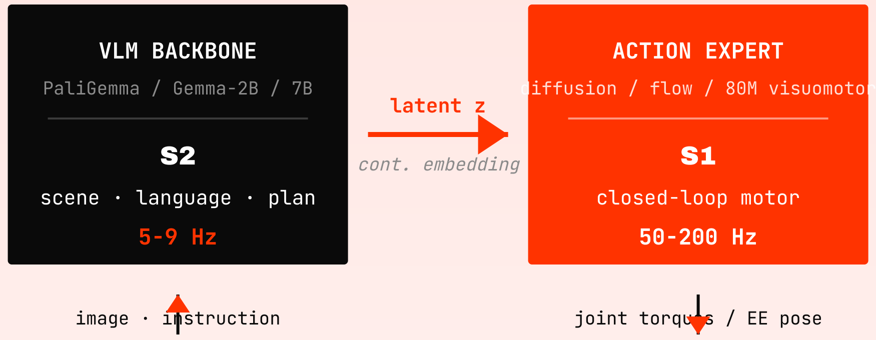
HUMAN · Kahneman 2002



	SYSTEM 1	SYSTEM 2
speed	fast · automatic	slow · deliberate
effort	parallel · effortless	serial · effortful
example	*driving on empty road*	*17 × 24*

SAME PATTERN

ROBOT · Helix · GR00T · π0

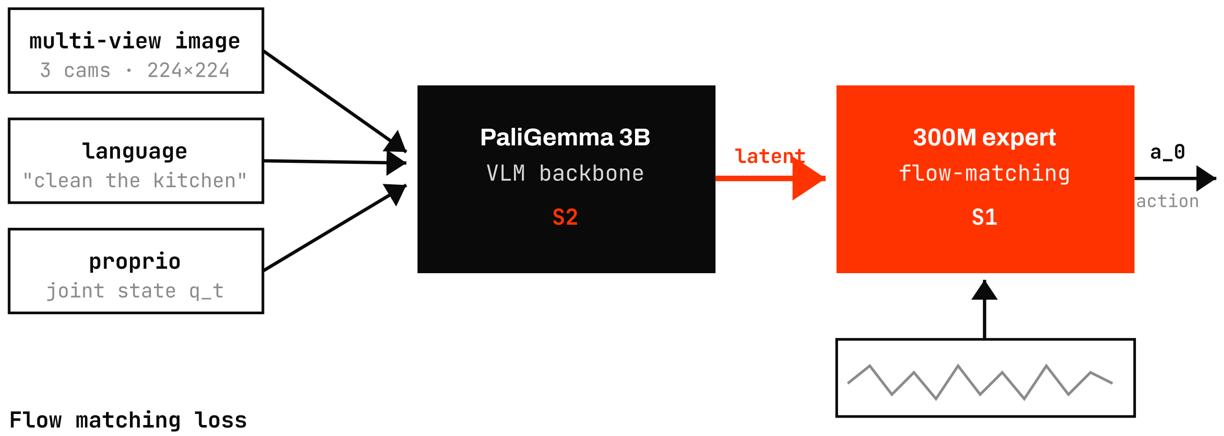


MODEL	S2	S1
Helix	7B VLM @ 7-9 Hz	80M @ 200 Hz
π0	PaliGemma 3B	300M flow expert
GR00T N1	VLM planner	diffusion transformer

Kahneman 2002 Nobel Lecture · Helix figure.ai/news/helix · π0 arxiv:2410.24164 · GR00T N1 arxiv:2503.14734

π0 & π0.5 — flow-matching action expert in unseen homes

Π0 ARCHITECTURE (BLACK ET AL. 2024) · REDRAWN



a_t (noisy action chunk)

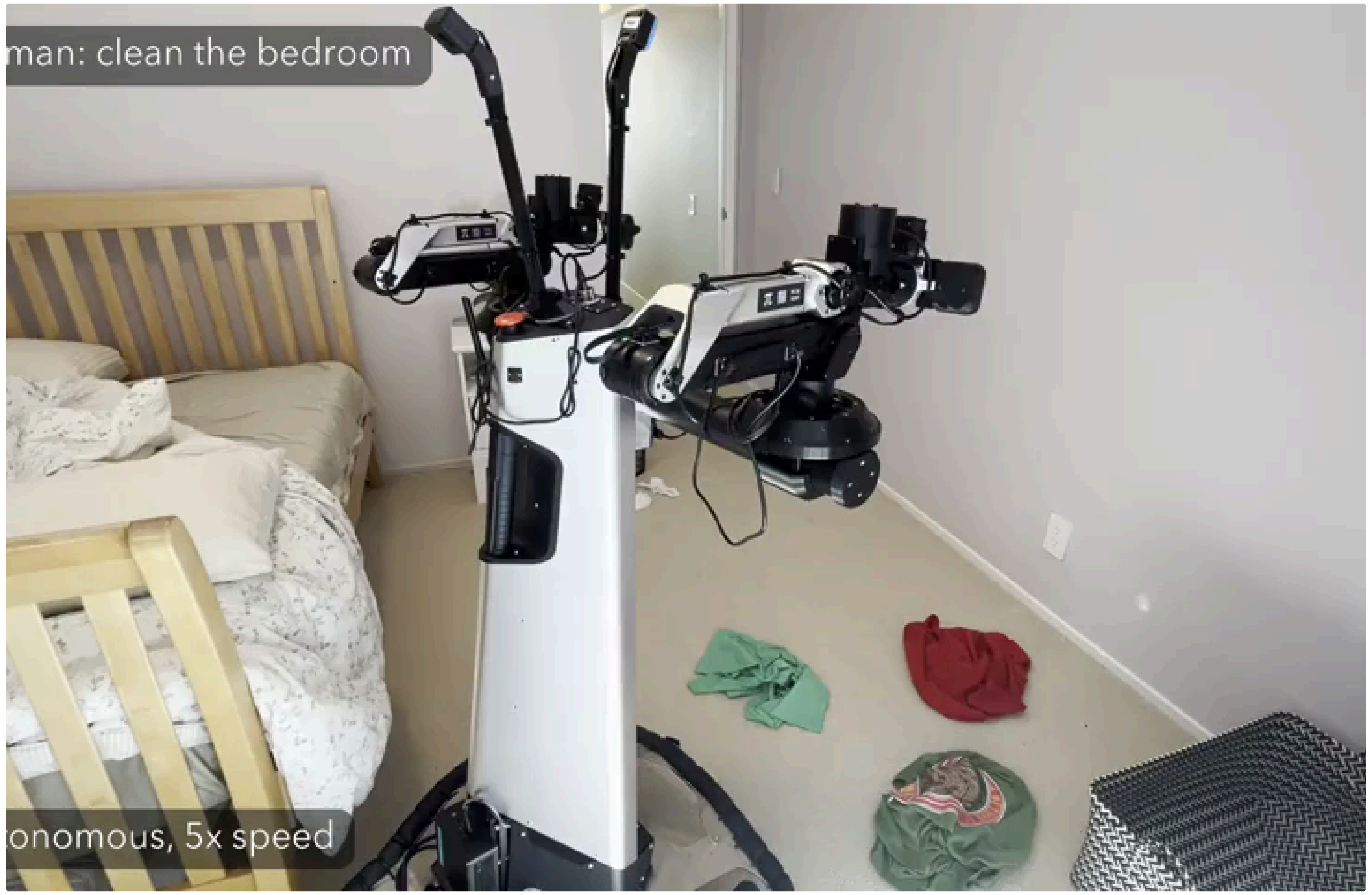
Flow matching loss

$L = \| v_\theta(a_t, t, z) - (a_1 - a_0) \|^2$

straight-line velocity field · deterministic ODE inference (10× faster than DDPM)

> Same dual pattern: VLM → latent → flow expert. Flow matching replaces DDPM's iterative noise prediction with a single learned velocity field — one short ODE solve at inference.

Π0.5 · NEVER-SEEN HOMES



FULLY AUTONOMOUS AIRBNB BEDROOM CLEANUP · 5× SPEED

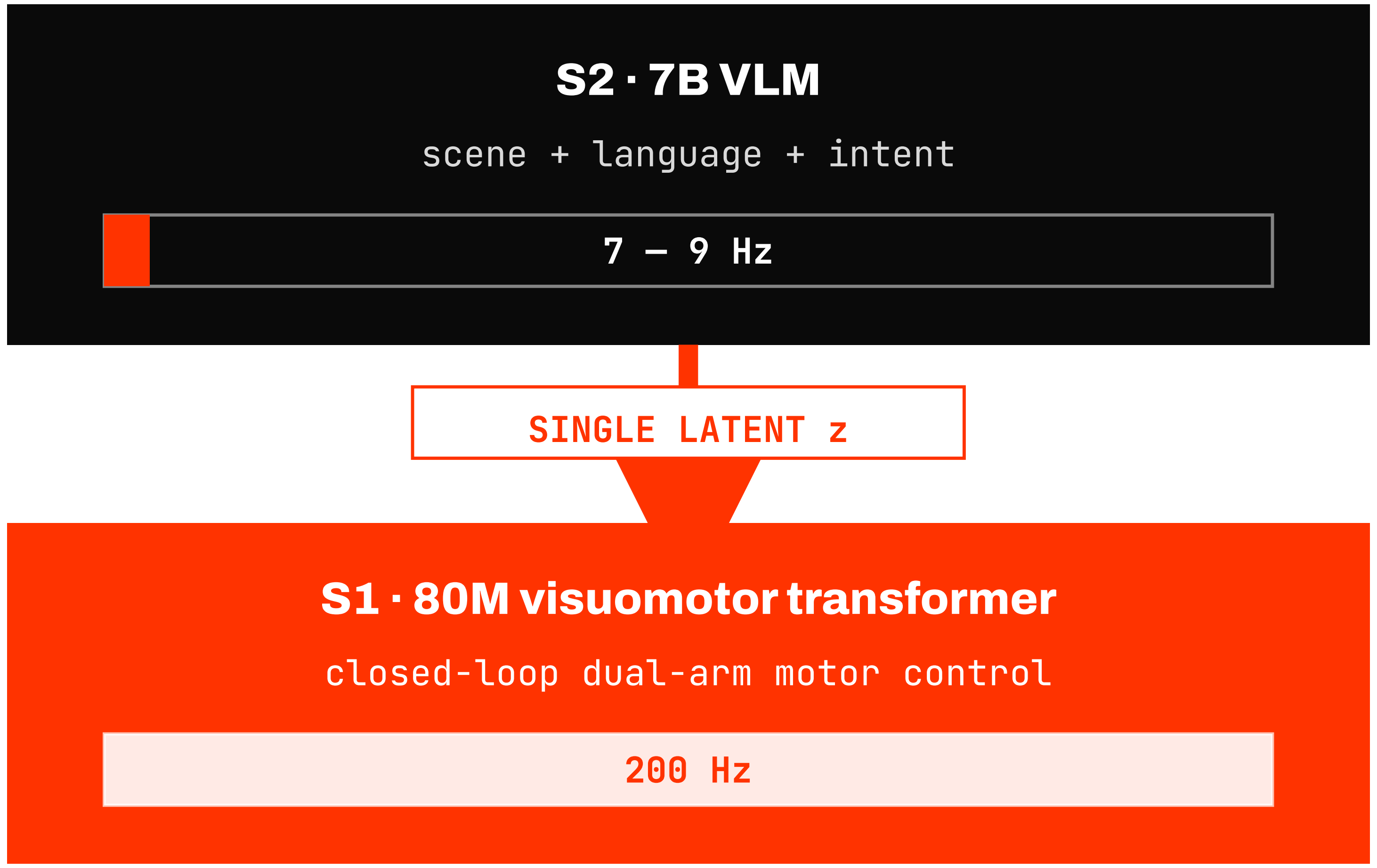
94% LANGUAGE-FOLLOW RATE, UNSEEN HOMES	**100+** NEVER-SEEN EVAL ENVIRONMENTS	**10×** INFERENCE SPEEDUP OVER DP-STYLE DDPM

Helix — Figure AI's 7B over 80M, two robots, one weights

TWO ROBOTS · FULL UPPER-BODY 35-DOF · SHARED WEIGHTS

HELIX GROCERY PUT-AWAY — FIGURE 02 HUMANOIDS COLLABORATE · FEB 2025

FREQUENCY SPLIT · BUILT FROM HELIX BLOG TEXT



~25× faster than S2 · bandwidth makes the difference

SINGLE WEIGHTS

Both robots run **the same network** — no role-specific finetuning, no per-robot models.

ON-BOARD INFERENCE

Both S1 and S2 run on an **embedded GPU** on the robot — no cloud.

CLOSED-SOURCE

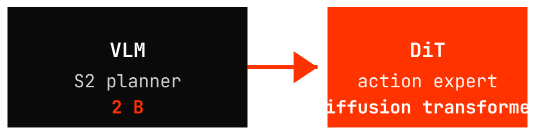
No paper, no weights — the architecture description is from the **blog text**. Figure 03 follow-up streamed live in May 2026.

GR00T N1 → N1.5 → N2 — one company, one year, three generations

MAR 2025 · GEN 1

N1

FIRST OPEN HUMANOID STACK



data: real + sim + human video

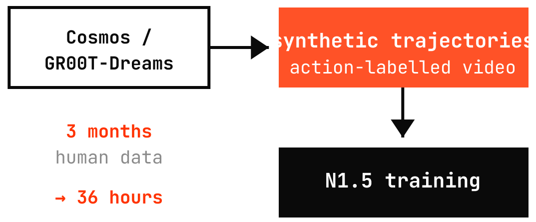
- S2 VLM + S1 diffusion transformer — the GR00T template
- Trained on heterogeneous data pyramid (real teleop, sim rollouts, ego-video)
- **Open weights** — first generalist humanoid policy you can download

arxiv:2503.14734

SEP 2025 · GEN 1.5

N1.5

SYNTHETIC DATA VIA GR00T-DREAMS



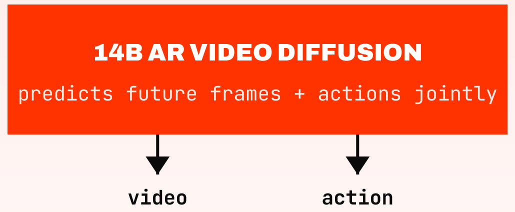
- Same dual-system architecture, supercharged with world-model-generated data
- Novel-object generalization — pick objects never seen in teleop
- Deployed on Unitree G1; canonical Ch3 → Ch6 bridge (synthetic data factory)

research.nvidia.com/labs/gear/gr00t-n1_5

FEB 2026 · GEN 2 (DreamZero)

N2 · a.k.a. DreamZero

VIDEO × ACTION JOINT GENERATION



- Dual-system collapses into one — **generation = policy**
- 7 Hz closed-loop @ 14B params via inference optimization
- Cross-embodiment in 30 min of plays — we revisit in **Ch 7**

arxiv:2602.15922 · dreamzero0.github.io

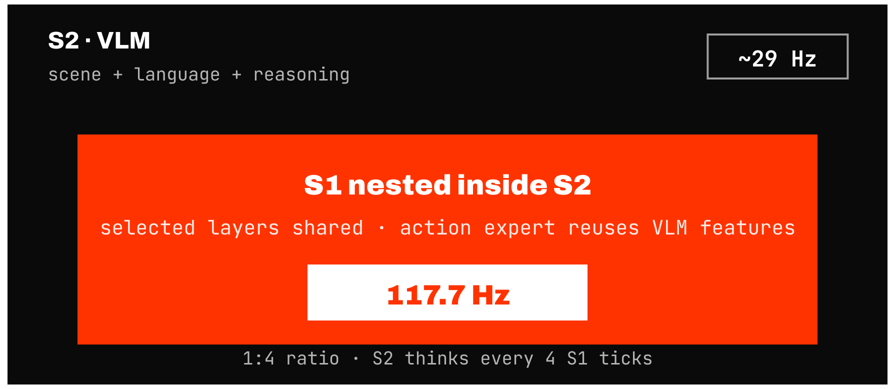
One company, one year — **open dual-system → world-model-generated data → joint video×action generation**. Train cost on synthetic data fell from **3 months → 36 hours**.

2026 frontier — Fast-in-Slow + Gemini Robotics 1.5

Fast-in-Slow

share parameters

FIS-VLA · CHEN ET AL. NEURIPS 2025



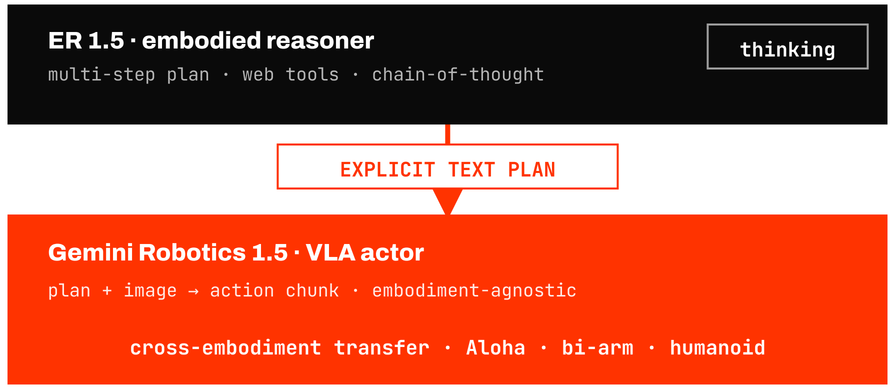
- **No latent vector handoff** — S1 reuses S2's intermediate features directly
- Slow loop runs at **1:4 ratio** relative to fast loop — one VLM forward per four action steps
- Reaches **117.7 Hz** on a single NVIDIA 4090 with chunk size 8

arxiv:2506.01953 · fast-in-slow.github.io

Gemini Robotics 1.5

externalize reasoning

GOOGLE DEEPMIND · SEPT 2025 + 2026 FOLLOW-UPS



- S2/S1 split is **two models**: ER 1.5 (planner) drives Gemini Robotics 1.5 (actor)
- Communication channel is **natural language** — not a latent vector
- Plans are inspectable, debuggable, and use external tools (search, calculator)

deepmind.google/blog/gemini-robotics-15

2026 TREND

Beyond *separate* S1 & S2 (Helix, GR00T N1) — **share parameters** (FiS-VLA) or **externalize the reasoning channel** (Gemini Robotics 1.5). Two opposite answers to "make the bridge thicker."

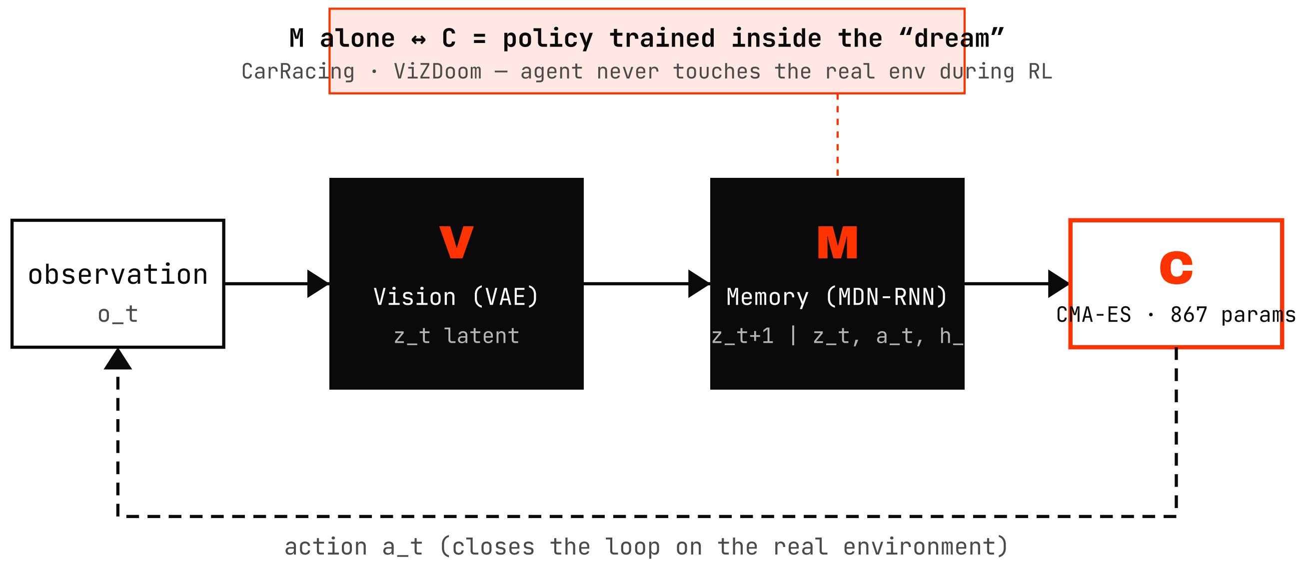
06

World Models.

Can a model learn the dynamics of the world — and use them?

Origin — Ha & Schmidhuber 2018

THE V → M → C DECOMPOSITION



A world model is a latent simulator of environment dynamics.

- **V** — compress pixels to z
- **M** — predict next z given action
- **C** — tiny controller, trained *inside* M's rollout

The original pitch: *train the agent in its own dream.* Every WM since — Dreamer, V-JEPA, Cosmos, DreamZero — is a re-mix of these three boxes.

The Dreamer family — one idea, five years

HAFNER ET AL. · LATENT-IMAGINATION POLICY LEARNING

2020 · ARXIV:1912.01603

Dreamer V1

RSSM latent dynamics + actor-critic in imagination. **Continuous control SOTA** (DMC).

2021 · ARXIV:2010.02193

Dreamer V2

Discrete latents. First model-based agent to **beat humans on Atari 200M**.

2023 · ARXIV:2301.04104

Dreamer V3

One set of hyperparams across **150+ tasks**. First to collect **Minecraft diamond** from scratch.

HISTORICAL ANCHOR · CORL 2023

DayDreamer — Dreamer on real robots

Wu et al. trained a real quadruped to walk from scratch in **~1 hour** — physical-world WM-based RL, no sim. The earliest concrete bridge from V/M/C to robotics.

World model = imagination simulator the policy can train inside.

- One family, three generations — the same V·M·C recipe, scaled
- RL framing dominated until 2023
- DayDreamer proves it transfers to a *physical* robot — bridge to Ch 6.10 (Slide 61)

Two axes — methodology × purpose

SAME WORD, DIFFERENT CELLS

	ENTERTAINMENT / GAMES	ROBOTICS / PHYSICAL AI
LATENT DYNAMICS (RNN/RSSM)	**Dreamer V3 (Minecraft)** 2023 diamond from scratch — in-game RL	**DayDreamer** CoRL '23 quadruped, 1 h real-world walk
JEPA (LATENT PREDICT)	— not a games line	**V-JEPA 2 · LeWorldModel · DINO-WM** 2025-26 action-conditioned representation prediction (Slides 53-54)
VIDEO GEN (PIXELS)	**Genie 1 · 2 · 3 · Oasis · Hunyuan-GameCraft · Mirage · MS Muse / WHAMM** 2024-26 playable game worlds at 10-24 fps	**Sora 2 · Veo 3 · Cosmos Predict · WAN 2.5** 2025-26 video gen as *physics simulator* (Slides 55-56)
3D / SPATIAL (GEOMETRY)	**Genie 3 (long-horizon consistency)** 2025 blurs into the robotics column —	**4D-GS · Cosmos Transfer · Lyra 2 · VGGT / VGGT-Ω · MapAnything** 2025-26 3D scenes + geometric foundation models (Slides 57-59)

Today we focus on the right column. The left column is moving fast too — and its data · architecture often crosses over (Genie 3 → sim asset, GameCraft engines → robot synth).

NEW · deepmind.google/blog/genie-3 · oasis-model.github.io · microsoft.com/research/.../muse · hunyuan-gamecraft

Three modern (robotics-relevant) families

What "WM" actually points to in 2025-26 robotics papers — three branches.

FAMILY 1

JEPA · latent prediction

predict *representations*, not pixels

- V-JEPA 2 (Meta, 2025)
- LeWorldModel (2026)
- DINO-WM (ICLR 2025)

Bet: abstraction & efficiency. Don't waste compute reconstructing pixels you'll never plan over.

→ Slides 53-54

FAMILY 2

Video gen as simulator

pixel-perfect rollouts

- Sora 2 (OpenAI)
- Veo 3 / 3.1 (DeepMind)
- Cosmos Predict 1 · 2.5 (NVIDIA)
- WAN 2.5 (Alibaba) · Genie 3

Bet: if you can generate the next 5 seconds correctly, you have a simulator — for data, eval, and rollout.

→ Slides 55-56

FAMILY 3

3D & spatial WMs

explicit geometry

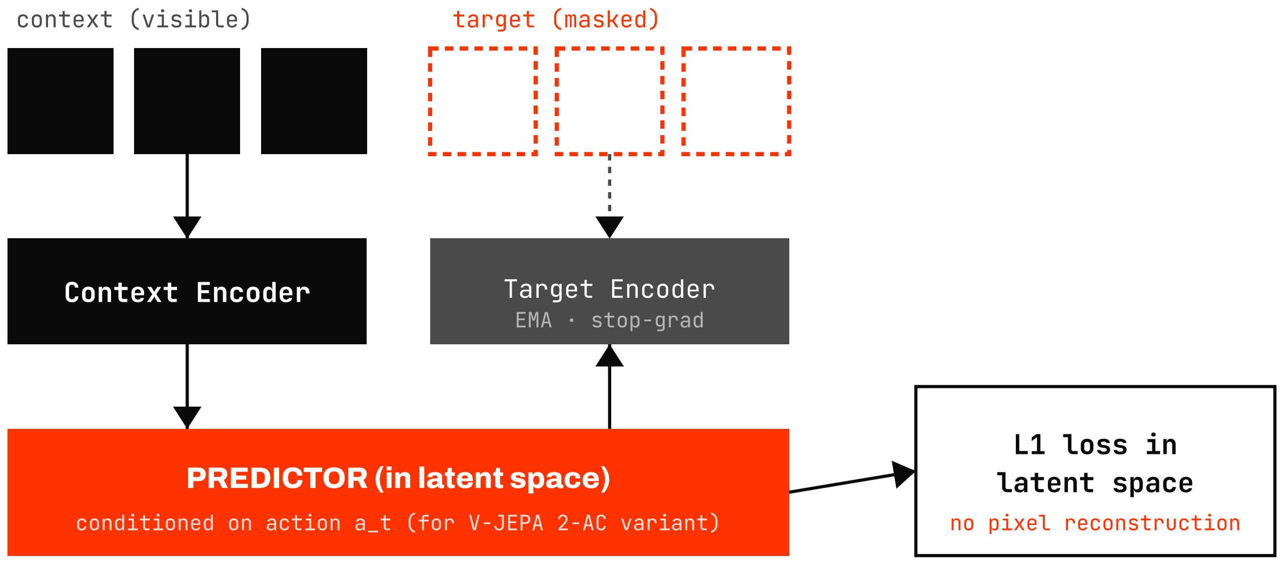
- 4D Gaussian Splatting
- Cosmos Transfer
- Lyra 2 (NVIDIA, 2026)
- VGGT / VGGT-Ω / MapAnything

Bet: robots live in 3D. Carry geometry, don't re-infer it every frame.

→ Slides 57-59

Family 1 · V-JEPA 2 (Meta, 2025)

PREDICT MASKED *REPRESENTATIONS*, NOT PIXELS — 1.2 B PARAMS



Downstream → action understanding (Something-Something v2) · robot manipulation

"Don't predict pixels you'll never plan over."

- 1.2 B params · trained on internet video
- SOTA on action understanding & anticipation
- Action-conditioned variant for manipulation

> LeCun's long-standing pitch finally shipped at scale — representation prediction beats pixel prediction on efficiency.

Family 1 · 2026 JEPA-line

After V-JEPA 2 — cheaper, end-to-end, plannable.

MAR 2026 · ARXIV:2603.19312

Maes · LeCun · Balestriero

LeWorldModel

Stable end-to-end JEPA *from raw pixels*. No frozen encoder. **15 M params** total — orders of magnitude smaller than V-JEPA 2.

48× FASTER PLANNING

15M TOTAL PARAMS

github.com/lucas-maes/le-wm

ICLR 2025 · ARXIV:2411.04983

NYU · Meta

DINO-WM

World model built on top of frozen **DINOv2** features — *zero-shot planning* by gradient descent over latent trajectories. Mazes, pushing, navigation.

> Showcase: a strong pretrained visual encoder + a small dynamics head is enough to plan over.

dino-wm.github.io

Common direction: **small, plannable, end-to-end**. JEPA line is now the cheapest WM-for-planning option — if you don't need pixels back.

Family 2 · Video gen as simulator

IF YOU CAN GENERATE THE NEXT 5 S CORRECTLY, YOU HAVE A SIM.

OPENAI · 2024-10 · V2 2025-10

Sora 2

Text-to-video w/ native audio. *Consumer app shut Apr 2026 — API only.*

DEEPMIND · V3.1 2025

Veo 3 / 3.1

1080p / 4K, 8 s clip, native audio & dialogue. Frames-to-video for control.

NVIDIA · 2025-01 · ARXIV:2501.03575

Cosmos Predict

Physical-AI WFM — pretrained for sim-to-real & robot data gen.

DEEPMIND · 2025-08

Genie 3

24 fps / 720p / multi-minute consistency — long-horizon scene memory.

Historical anchor · ICLR'24 Outstanding · [ARXIV:2310.06114] **UniSim** (Yilun Du et al.) — first paper to call video generation an *interactive real-world simulator*. Everything in this slide is downstream of that framing.

Family 2 · 2026 — video gen WMs grow up

2025 = ARRIVED · 2026 = BECOMES A SIM

NVIDIA ARXIV:2511.00062

Cosmos Predict 2.5

Unified T2W / I2W / V2W flow model · **200 M clips** · 2 B + 14 B. Targets AV & robotics.

DEEPMIND 2025 DEMO

Veo 3 · robotics

Veo simulator used by Gemini Robotics for eval (see Slide 61 — 1,600+ real evals).

ALIBABA 2025 · NO ARXIV

WAN 2.5

Open-source
video gen baseline
backbone for Lyra 2 (Slide 58)

Weights/code track on Wan-Video/Wan2.2. No formal paper — release notes only.

TENCENT / KUAISHOU 2025-11

GameCraft 2 · Kling 2

Instruction-following
game WM
text + kbd + mouse

Hunyuan-GameCraft v2 (arXiv:2511.23429) · the games ↔ robotics bleed continues.

One-year jumps on **four axes at once** — resolution · clip length · physical consistency · action conditioning. The category renamed itself from "video gen" to "neural simulator".

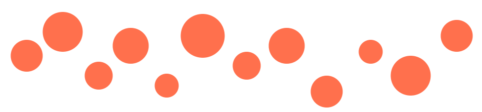
NEW · [ARXIV:2511.00062 / 2511.23429] · github.com/Wan-Video/Wan2.2 · deepmind.google/blog/gemini-robotics-1-5

Family 3 · 3D & spatial world models

Carry the geometry — don't re-infer it every frame.

CVPR 2024 · ARXIV:2310.08528 HUST (Wu et al.)

4D Gaussian Splatting



3D Gaussians + per-Gaussian motion over time t
real-time photoreal renders of dynamic scenes

- Explicit 3D Gaussians, not implicit fields
- Real-time render · novel-view eval

The geometric substrate Lyra 2 (Slide 58) generates and Cosmos Transfer renders on.

MAR 2025 · ARXIV:2503.14492 NVIDIA

Cosmos Transfer 1

- Depth / seg / edge → photoreal video
- Closes the sim-to-real gap on policy rollouts

Sim engines do geometry, Cosmos Transfer does pixels — the standard 2025-26 robot data pipe.

Family 3 · NVIDIA Lyra 2

One image / prompt → navigable 3D Gaussian scene — a *geometry-native* WM.

LYRA 2.0 TEASER — TEXT/IMAGE → 3D GAUSSIAN SPLAT SCENE

A scene-generation foundation model.

- Built on **WAN 2.1** video backbone — distilled into 3D-GS
- Output is *3D Gaussian splats*, not video — truly navigable
- Plugs straight into Isaac Sim for robot rollouts

NEW · [ARXIV:2604.13036] · "Lyra 2.0: Explorable Generative 3D Worlds" · NVIDIA SIL · Apr 2026

Family 3 · Geometric Foundation Models

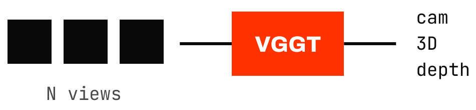
"World model" = *3D reconstruction itself* as a feed-forward foundation. The geometric backbone.

CVPR 2025 · BEST PAPER

arXiv:2503.11651

VGGT

Visual Geometry Grounded Transformer. **Feed-forward** 3D — cameras, depth, point map in one pass.



FAIR + Oxford

CVPR 2026 ORAL

arXiv:2605.15195

VGGT-Ω

VGGT scaled and shrunk. The 2026 update.

30%	15×	+77%
MEMORY	DATA	SINTEL

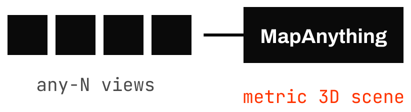
Wang · Vedaldi et al.

3DV 2026

arXiv:2509.13414

MapAnything

Meta + CMU. Universal feed-forward **metric** 3D — outputs scenes in real-world units, not just relative scale.



map-anything.github.io

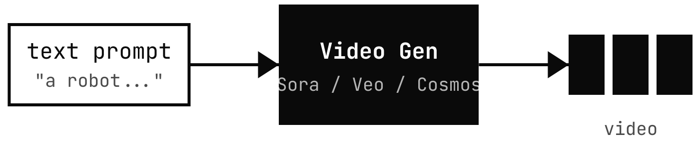
JEPA carries *latents*. Video gen carries *pixels*. This line carries **geometry itself** — what WoRV uses today for data & eval.

Action-conditioned **world models**

Without an action input, it's just video gen. *With* one — it's a robotics WM. This is the bridge to Ch 7.

GENERIC VIDEO GENERATION

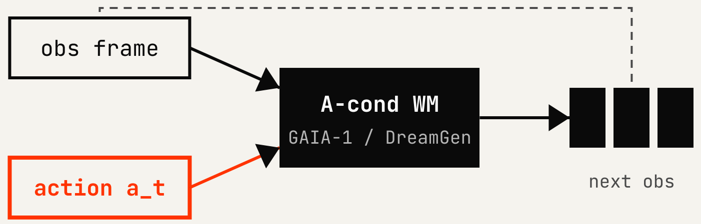
prompt → video



- One-shot prompt → one rollout
- No way to *steer* a robot through it
- Great for content. Not a policy interface.

ACTION-CONDITIONED WM

(obs, action_t) → obs_{t+1}



- Counterfactual rollouts — "what if I push left?"
- **GAIA-1** (Wayve, 9B, driving) · **GR00T-Dreams / DreamGen** (NVIDIA, arXiv:2505.12705)
- This is what makes WMs *useful for policies*.

Bridge to Ch 7: once you can condition on action, you can also generate it. That's a **WAM** (World Action Model) — DreamZero (Slide 66).

Why WMs matter for robotics

Three tools in one model — that's why everyone is racing here.

1 DATA

Synthetic data factory

A trained WM is a generator of new (obs, action) pairs. GR00T-Dreams cut training from **3 months → 36 h** by replacing teleop with WM rollouts.

See Ch 3 Slide 28 · arXiv:2505.12705

2 EVAL

Zero-shot evaluation

Score a policy by rolling it out *inside* the WM — no real-robot time. DeepMind's "Evaluating Gemini Robotics in Veo World Simulator" ran **1,600+ real evals** this way; academia's **WorldEval** (Midea) does the same.

1,600+
REAL EVALS (VEO)

WorldEval · Gemini-in-Veo (DeepMind)

3 POLICY

Policy via rollout

Plan or RL *inside* the WM — the original Ha & Schmidhuber dream. DayDreamer (CoRL'23) on a real quadruped in 1 h; Dreamer V3 on Minecraft.

Dreamer V3 · DayDreamer · → Ch 7 WAMs

Data · Eval · Policy — three of the most expensive things in robotics, all addressed by one trained model. That's why this chapter is the *longest* in the talk.

07

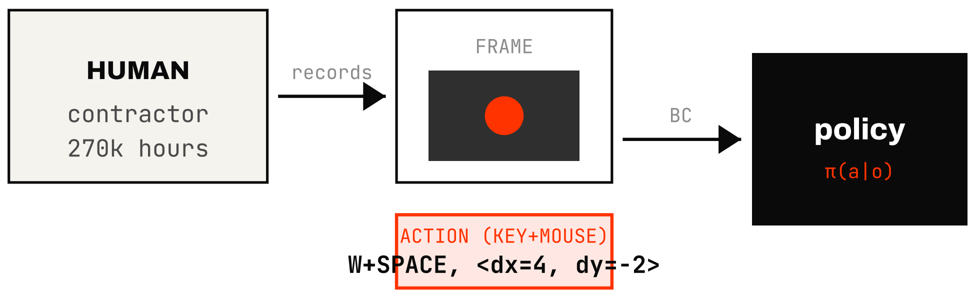
World Action Models.

Does knowing the world's dynamics turn into ***zero-shot policies****?*

The **IDM** idea — label video *backwards*

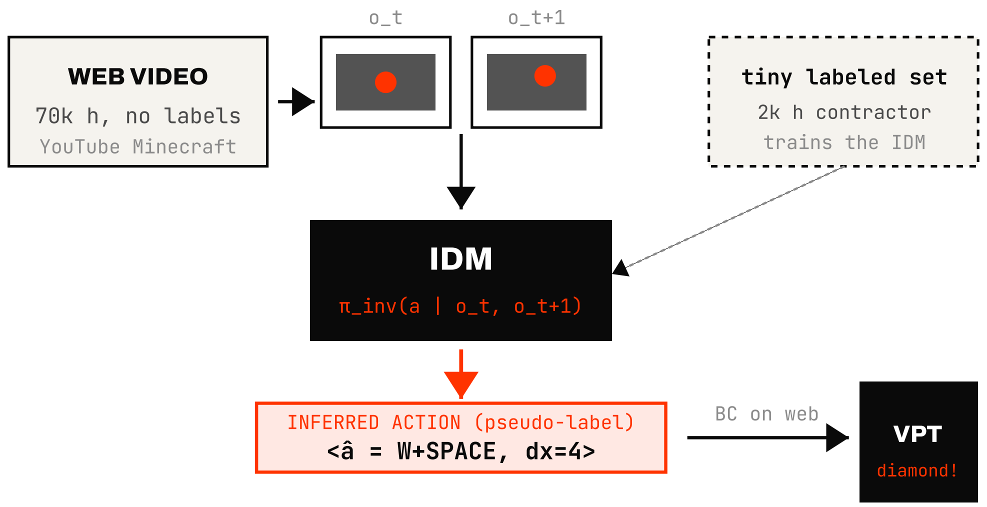
BAKER ET AL. · OPENAI VPT · 2022

(A) SUPERVISED — TELEOP / CONTRACTOR



Cost: human pays per hour. Caps at $\sim 10^5$ frames.

(B) IDM — INFER ACTION FROM 2 FRAMES



The trick: a small labeled set teaches the IDM → the IDM *labels* 70k h of unlabeled web video → behavior-clone the labeled web video. First *foundation* policy in Minecraft. The same recipe ports to robotics — `UniPi, GR-1, VPP, NovaFlow` (talking-point only).

LAPA — latent actions from video

YE ET AL. · ICLR 2025

LAPA HERO TEASER · LATENTACTIONPRETRAINING.GITHUB.IO

(1) HUMAN VIDEO
VQ-VAE
on inter-frame
→ discrete latent z

→

(2) PRETRAIN
Latent VLA
predict next z
no robot data

→

(3) FINETUNE
align to real a
tiny robot dataset
+6.22% over OpenVLA

30× more pretraining-efficient than action-labeled VLA.

🇰🇷 KOREA · KAIST-LED

4 of the first-line authors — and both advisors — are at KAIST.

Seonghyeon Ye · first author	KAIST
Byeongguk Jeon	KAIST
Sejune Joo	KAIST
Joel Jang	UW
Kimin Lee · equal advising	KAIST
Minjoon Seo · equal advising	KAIST

- Discrete **latent action** — learned, not annotated
- Pretrain on human / web video, finetune on small robot set
- Strongest *made-in-Korea* entry in the WAM lineage

From IDM / LAPA to joint video × action modelling

STEP 1 · 2022

IDM

`video → action`

Infer the action that bridged two frames. *Recover* the missing label.

VPT · UniPi · GR-1

→

STEP 2 · 2024

LAPA

`video → latent z → action`

Drop the hand-defined label. *Learn* the latent action with VQ-VAE.

🇰🇷 KAIST

→

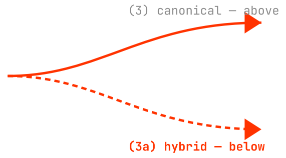
STEP 3 · 2026

WAM

`video ↔ action (joint)`

Stop separating them. Generate the future video *and* the action together.

DreamZero (canonical)



PARALLEL EMERGENCE

STEP 3A · SAME ERA

VLA + WM hybrid

`VLA loss + video-pred loss · one backbone`

Don't fully *generate* the future — just *predict* it as an auxiliary objective alongside the action head.

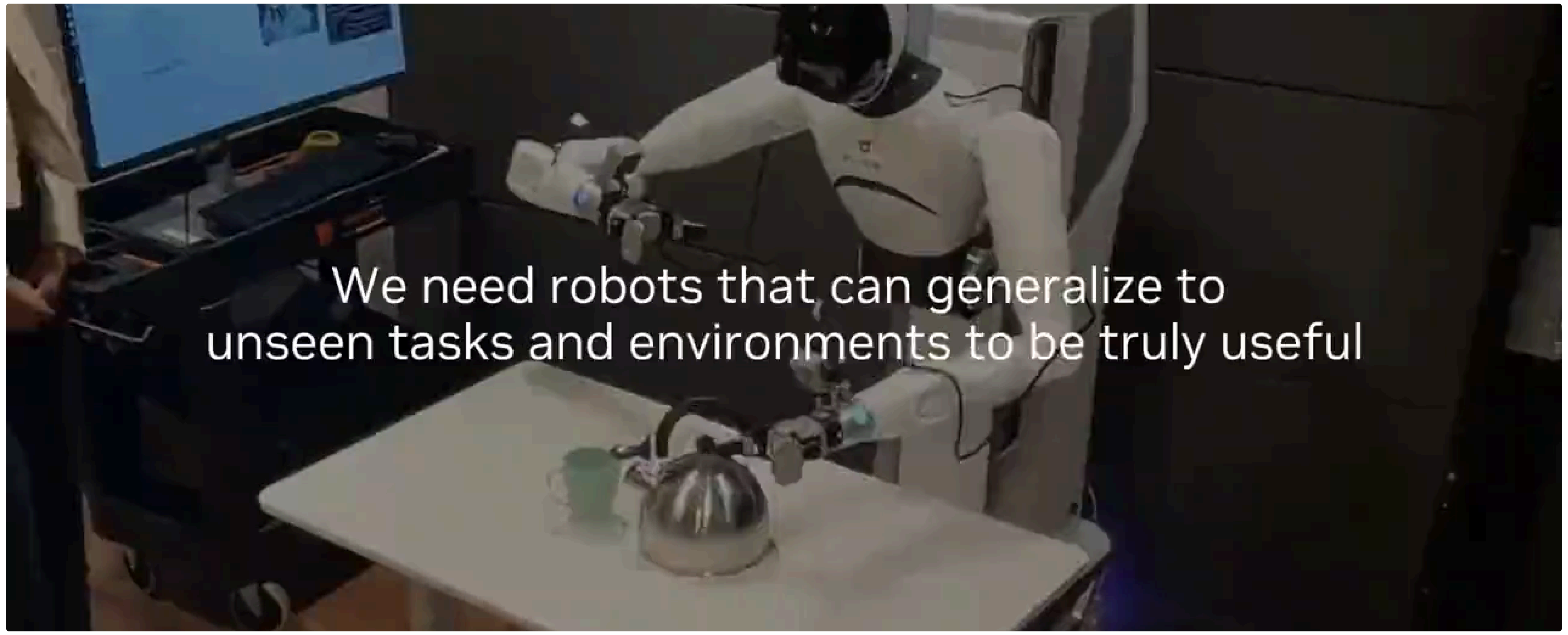
InternVLA-A1 · LingBot-VA · 12+ more (slide 68)

> Two ways to *not separate* video and action. Slide 66 covers the canonical line (DreamZero); slide 68 covers the hybrid line.

arXiv: 2206.11795 · 2410.11758 · 2602.15922 · 2601.02456

DreamZero — “World Action Models are Zero-shot Policies”

NVIDIA GEAR · 2026-02



DREAMZERO OVERVIEW · 14B AR VIDEO DIFFUSION + ACTION

62.2%
AGIBOT · SEEN
vs VLA 27.4%

39.5%
AGIBOT · UNSEEN
vs from-scratch ≈0%

49%
DROID · UNSEEN VERBS
vs π0.5 33% · GROOT N1.6 31%

Why-it-works: VLA = semantic prior · WAM = physical prior. Generating the future video *conditions* the action on a real rollout.

SINGLE-BACKBONE VIDEO × ACTION LINEAGE · 1 YEAR, 6 STEPS

- 2024 · ICLR — **GR-1** — ByteDance →
- 2025 · ICML — **VPP** — Tsinghua · UCB →
- 2025 · RSS — **UVA** — Stanford →
- 2025 · RSS — **UWM** — UW WEIRD →
- 2026 · ICLR — **GE-Act** — AgiBot →
- 2026-02 · arXiv — **DreamZero** — NVIDIA · 14B

DreamZero didn’t appear out of nowhere — a year+ of single-backbone work made it scalable.

DreamZero deployment — 7 Hz at 14B · 30 min to a new embodiment

(A) SYSTEM · RUNNING A 14B WAM AT ROBOT RATE

14B AR video diffusion + action
multi-step denoise

distill →

DreamZero-Flash
single-step variant
causal cache · KV reuse

→

7 Hz
control rate

CONTEXT · control freq for big VLMs

RT-2 · 1 Hz — OpenVLA · 5 — DreamZero · 7 — π0 · 50 — Helix S1 · 200

14B normally implies *seconds-per-step*. Flash distillation + KV-cache reuse lands at **robot-usable rate** — not in the π0 zone, but enough for tabletop manip.

(B) CROSS-EMBODIMENT · 30 MIN ON YAM

YAM TEDDY TRANSFER · 30 MIN OF PLAYS, NEW ROBOT · ONE MODEL

Model + deployment, advancing together.

The same generative prior that gives zero-shot tasks also gives *fast embodiment adaptation*.

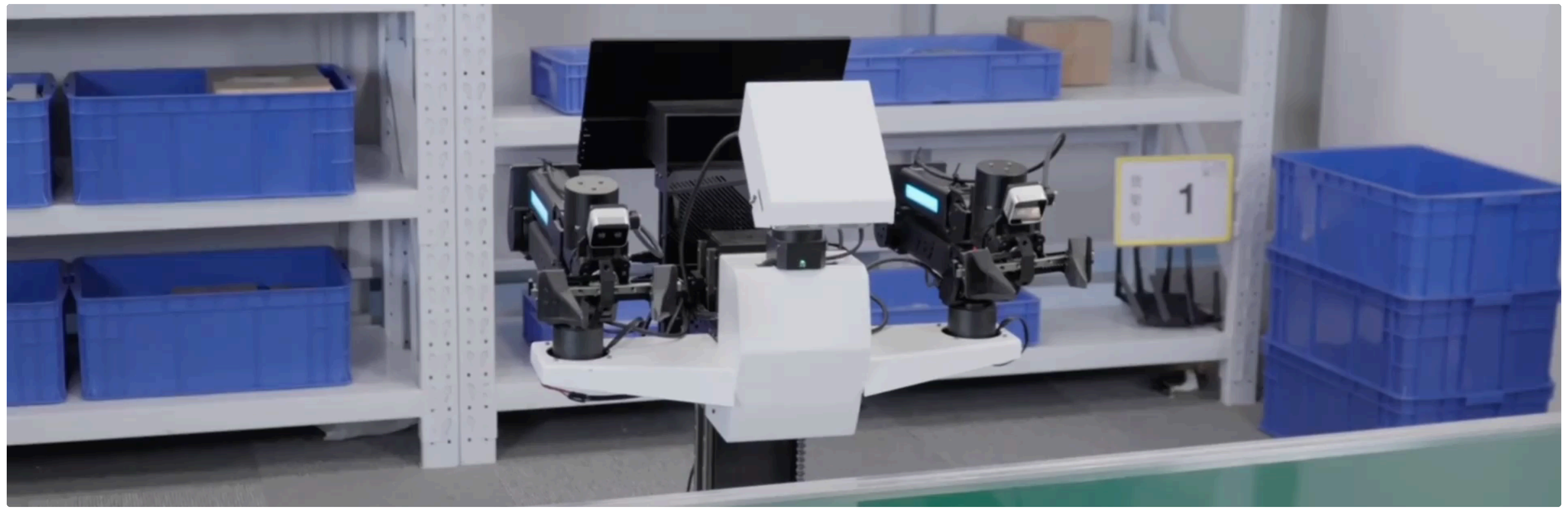
Concurrent VLA × WM hybrids — early 2026

SAME INTUITION · DIFFERENT RECIPES

InternVLA-A1

2026-01-05

Shanghai AI Lab + Humanoid Robot (Shanghai) Co. · 42 authors · lead Jia Zeng / Jiangmiao Pang



+26.7% DYNAMIC TASKS

75.1% AVG / 12 REAL TASKS

MoT (Mixture-of-Transformers) — one backbone, three experts: *understanding*, *generation*, *action*. 2-3B params, 692M pretraining frames.

[ARXIV:2601.02456]

Causal World Modelling aka LingBot-VA

2026-01-29

Ant Group / Robbyant · Li / Zhang / Luo et al. · corresp. Yinghao Xu

FRAMEWORK

shared latent · video + action
AR-diffusion · async inference

paper Fig 1 · framework2.pdf

92.9% LIBERO · EASY

91.6% LIBERO · HARD

AR diffusion over future frames + policy execution in **shared latent space**. SOTA on all 6 real-robot tasks vs π0.5; +8-9% at horizon 3.

[ARXIV:2601.21998]

Same intuition, different design. Plus 12+ more in the NTU MARS survey · GE-Act · Motus (+45% over π0.5) · BagelVLA · FRAPPE · STARRY · WAV .

WAM follow-ups — mid 2026 · four directions in ~3 months

POST-DREAMZERO

SIMPLIFY ARCHITECTURE

2026-04-30

Being-H0.7

Peking U / BeingBeyond · Zongqing Lu

No future-video generation. Just a **dual branch with learnable latent queries** between perception and action. Pretrained on `200k h ego video + 15k h robot demos` — the Ch3 ego pool, finally cashed.

99.2%	62.1%	49.2%
LIBERO	ROBOCASA	GR1

[ARXIV:2605.00078 · ← CH3 FORWARD-REF]

TRANSFER EFFICIENCY

2026-05-07

CKT-WAM

Tsinghua + LivsynRobotics + Shanghai AI Lab

Teacher hidden states → compressors → routed adapters → **student text-embedding space**. Knowledge transfer between WAMs without touching the backbone.

86.1%	1.17%
LIBERO-PLUS	TRAINABLE PARAMS

[ARXIV:2605.06247]

OBJECT ADDRESSING

2026-05

OA-WAM

Object-Addressable WAM

Treat objects as first-class addressable entities inside the world model. Lets the policy refer to “the red cup” the way a VLA refers to a token — not a pixel patch.

97.8%
LIBERO

[ARXIV:2605.XXXXX]

COUNTER-NARRATIVE

DO WE NEED IMAGINATION?

2026-03

Fast-WAM

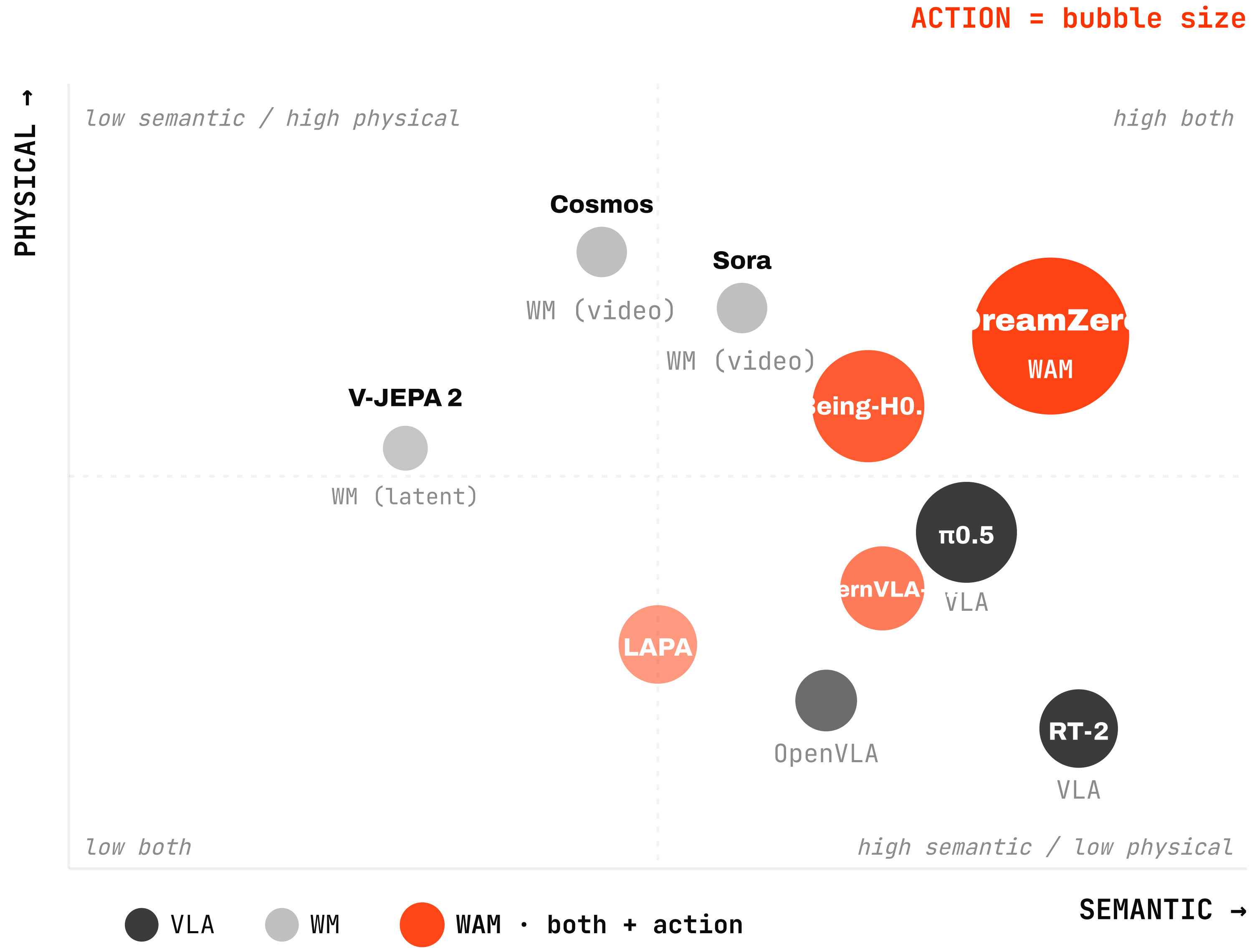
“*Test-time future imagination is unnecessary.*” Skip the rollout, get SOTA at **4× speed** — directly challenges DreamZero / Being-H0.7’s core premise.

190 ms	4×
INFERENCE	FASTER

FFDC-WAM threads the needle with **conditional imagination**.

WMs vs WAMs vs VLAs — 2026 taxonomy (3 axes)

SEMANTIC · PHYSICAL · ACTION



WAMs sit at the intersection.

- VLAs — **semantic** mostly, action small
- WMs — **physical** mostly, no action head
- WAMs — both, *and* action native

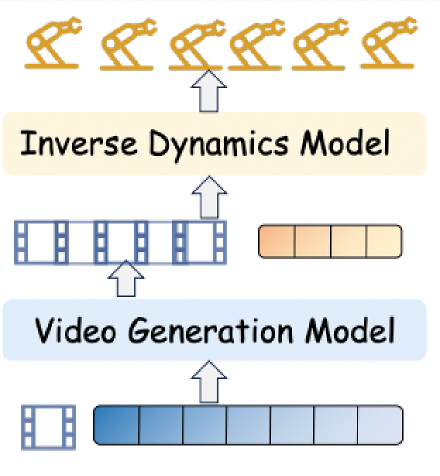
> The bubble size = how richly the model represents *action*. Sora · Cosmos · V-JEPA 2 have none.

WMs vs WAMs vs VLAs — 5-architecture view

NTU MARS SURVEY · ARXIV:2605.00080

PATTERN 1

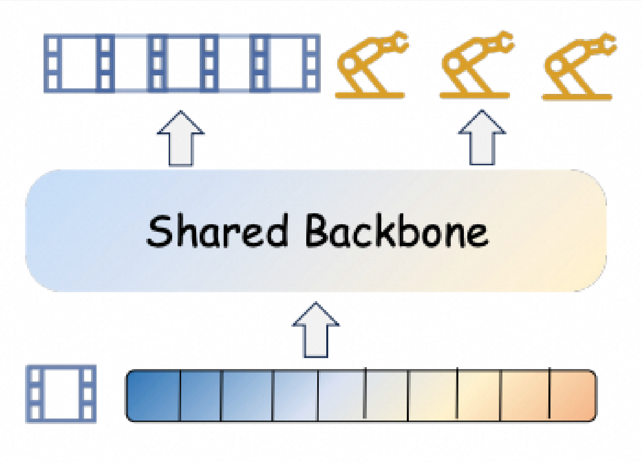
IDM-style



Decouple: *predict subgoal / latent*, then act. VPT · UniPi · LAPA

PATTERN 2

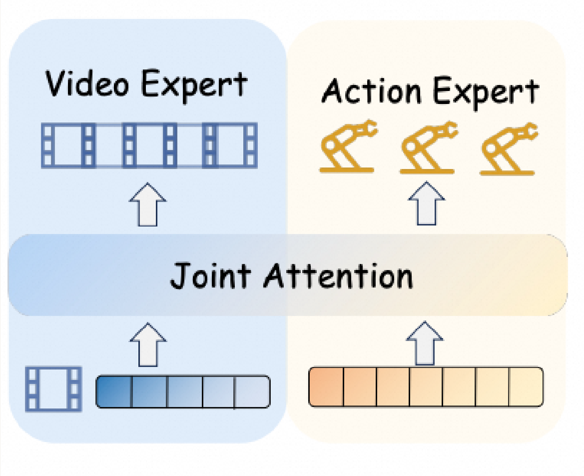
Single-backbone



One model, video × action **jointly**. UVA · UWM · DreamZero

PATTERN 3

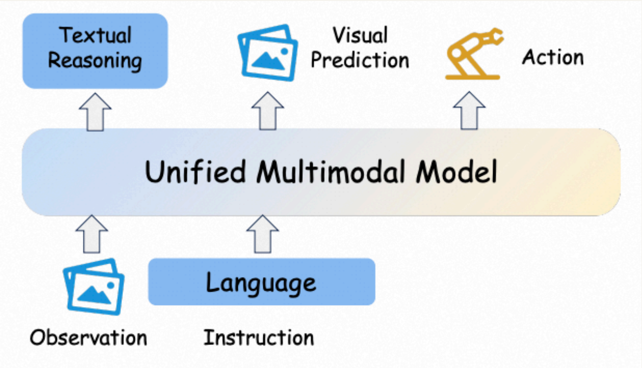
MoE / MoT



Expert fusion / joint attention. LingBot-VA · GE-Act · BagelVLA

PATTERN 4

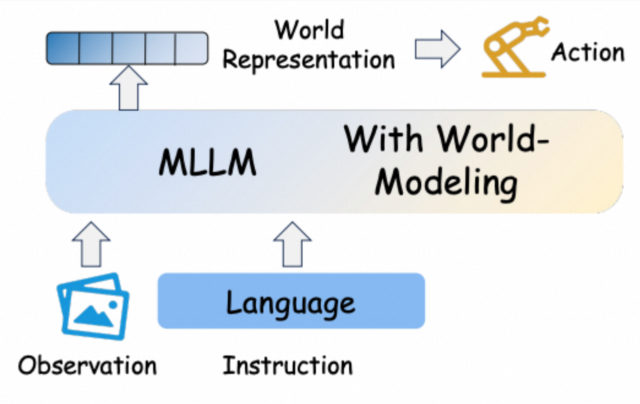
Unified-VLA



VLA + *foresight* / video pred. as aux. GR-1 · InternVLA-A1 · UniVLA

PATTERN 5

Latent-space



JEPA-style, *no* pixel gen. V-JEPA 2 · VLA-JEPA · FLARE · Being-H0.7

Every model in today's talk lives in one of these 5 boxes. Patterns 2 and 3 are where the action is in 2026 — canonical WAM (slide 66) and VLA × WM hybrid (slide 68).

08

WoRV @ maum.ai

How we're working on all of this from Korea

World model for Robotics and Vehicle control

World Robot Verse

maum.ai's physical-AI division.
We build **foundation models that work in the physical world**.

@ Pangyo IT Center #2   K-Humanoid Alliance member

OUR BET — 2026 TAXONOMY

We bet on the WAM × dual-system line.

VLA	WM	WAM
semantic-heavy	world-heavy	★ WoRV position

(Taxonomy reused from Ch7 slide 70 — we build a Korean stack on the same line.)

What we do — three stacks

PILLAR 01

Data Infrastructure

ALOHA / UMI / egocentric video pipelines + synthetic data factory. Every trick from Ch3, running in-house.

`own rigs + UMI variants + ego ingestion`

PILLAR 02 · LEAD

CostNav `arXiv:2511.20216`

A navigation benchmark measured in **real economic cost**. 23 co-authors; I'm the last author.

−$27.36
CANVAS / RUN

−$35.46
LIDAR+GPS / RUN

No baseline is economically viable yet — we turn an academic hypothesis into a cost number.

PILLAR 03

World Models & WAM

On top of Ch6 / Ch7 lines — in-house video-diffusion + joint action training. The actual implementation of our WAM × dual-system bet.

`internal · public release TBD`

> Public HuggingFace assets: `maum-ai/CostNav` · `CostNav-Teleop-Dataset` · `CostNav_baseline`

Deployment — 5 verticals

SPECIFIC CUSTOMERS ARE CONFIDENTIAL — VERTICALS ONLY

01
Agriculture

02
Construction

03
Maritime

04
Defense

The model runs in 5 real industries — that fact alone is the point. Specific partners and impact figures stay verbal — confidentiality policy.

We are **hiring** — people to solve this with us

POSITIONS

- 01 **Research Scientist** — VLA · WM · WAM
- 02 **Research Engineer** — data infra · training systems
- 03 **Research Associate · Intern** — undergrads welcome

`KR military alt-service: Research` `KR military alt-service: Industrial`

`@ Pangyo IT Center #2`

Hiring step 2 = **coding test + take-home assignment**. Take a swing. HUFS students: alternative-service tracks are on the table too.

[QR placeholder]
qrencode → Phase 8

recruitment.worv.maum.ai

worv_hr@maum.ai

Thank **you.**

Q & A
15 minutes

CONTACT

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WEB alohays.github.io
TEAM WoRV @ maum.ai (Pangyo)
HIRING recruitment.worv.maum.ai

FURTHER READING

"World Model for Robot Learning: A Comprehensive Survey"
Hou et al. · NTU MARS + Abbeel + Malik + Wu + Du · 2026-04
arXiv:2605.00080

Bookmark — for everything we didn't have time to cover.

[QR]	[QR]	[QR]
Deck	Survey	Hiring
alohays.github.io/ talks/hufs-2026-rfm	arXiv:2605.00080	recruitment. worv.maum.ai